

ANALYTIC CONSTRUCTION OF RATIONAL CURVES ON FANO MANIFOLDS

YUN-HENG DU, BIN GUO, AND SONG-YAN XIE

ABSTRACT. Inspired by methods for constructing entire curves in Oka geometry, we give an analytic construction of rational curves on a complex Fano manifold X . Yau’s theorem provides a Kähler metric with positive Ricci curvature. Using this curvature to guide deformations of holomorphic disks, we continue maps under a uniform area bound. The limiting process either produces a rational curve by bubbling or yields disks of radii tending to infinity whose principal limit retains a nonzero derivative normalization. In the latter case, the resulting nonconstant entire map $f : \mathbb{C} \rightarrow X$ has finite area and extends across infinity to a nonconstant holomorphic map $\mathbb{P}^1 \rightarrow X$. Combined with algebraic arguments, this principally analytic construction gives new proofs in characteristic zero of two known results: the rational connectedness of Fano manifolds, due to Campana and Kollár–Miyaoka–Mori, and Mori’s solution of Hartshorne’s conjecture on ample tangent bundles.

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1. INTRODUCTION AND MAIN RESULTS

1.1. The analytic construction problem

Our aim is to construct rational curves on complex Fano manifolds by analytic methods. A Fano manifold X is connected, smooth, and compact, with ample anticanonical line bundle K_X^{-1} . In his monograph *Ample subvarieties of algebraic varieties*, Hartshorne [27] asked whether a smooth complex projective variety with ample tangent bundle must be projective space. Mori’s solution [39] made the construction of rational curves a central tool of higher-dimensional algebraic geometry. His bend-and-break method combines deformation theory with reduction to positive characteristic and Frobenius.

The search for an analytic construction of rational curves has a long history. In 1980, Siu and Yau [45] used energy-minimizing harmonic spheres in their proof of the Frankel conjecture. Their argument uses positive holomorphic bisectional curvature, a stronger condition than the Ricci

2020 *Mathematics Subject Classification.* Primary 14J45, 32Q15; Secondary 14M22, 32Q10, 32Q45.

Key words and phrases. Fano manifold, rational curve, rational connectedness, holomorphic disk, Ricci curvature, Hartshorne’s conjecture.

positivity that Yau’s prescribed-Ricci theorem provides on every Fano manifold [48]. Already in 1982, Yau asked whether any two points of a compact Kähler manifold with positive Ricci curvature could be joined by a chain of rational curves [49, p. 44]. Rational connectedness of Fano manifolds was subsequently established by Campana and by Kollár, Miyaoka, and Mori [6, 35]. The question of constructing the curves by analytic methods remained a separate challenge. Siu later proposed approaching this problem through complex Monge–Ampère equations, without reduction to positive characteristic [44].

The folklore conjecture that every Fano manifold is Oka provides another motivation; see [20, p. 660] for this open problem. For the general framework of Oka theory and its relation to Stein geometry, we refer to Forstnerič’s monograph [19]. Alarcón and Forstnerič opened a new direction in Oka geometry by introducing Oka–1 manifolds and conjecturing, in particular, that every rationally connected projective manifold is Oka–1 [1, Conjecture 9.1]. Our recent proof of the projective statement [14, Corollary 1.8] brought new insights and methods for constructing entire curves. Guo and Xie had already suggested in [25, Section 5] that studying entire curves with slow growth could lead to analytic constructions of entire and rational curves on Fano manifolds. Our companion results motivate this approach, but the proof below neither invokes them nor assumes that the target is Oka–1.

Following the direction suggested by Guo and Xie [25], we pursue a precise growth condition: finite total area. The argument combines the Brody normalization and expanding-disk viewpoint of complex hyperbolicity with deformations of holomorphic maps from Oka geometry [46] and the compactness and removal principles for finite-area curves familiar from symplectic geometry. Positive Ricci curvature guides the deformations, keeping area uniformly bounded as the source radii tend to infinity. A nonconstant entire limit then extends across infinity to a rational curve; a suitable sphere may also appear earlier by bubbling.

A traditional Chinese pastry-making technique suggests a picture of the construction: a baozi is formed by gathering a disk of dough around its filling and closing the folds. Here the gathering recalls the deformation of holomorphic disks, while the closure recalls finite-area removal at infinity. Positive Ricci curvature provides the area control needed for this limiting step.

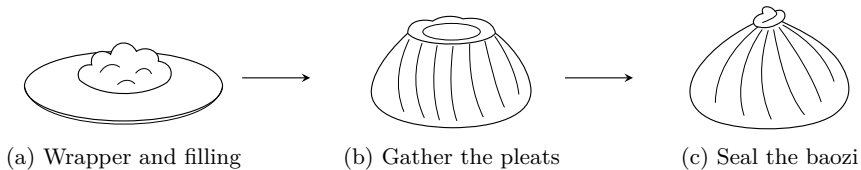


FIGURE 1. Shaping a baozi by gathering a dough wrapper around a filling and sealing the pleats. This traditional Chinese craft suggests an analogy for closure. In the proof, a finite-area entire principal map extends across infinity; bubbling is treated separately by marked compactness.

The first obstacle is to control area as the source radii tend to infinity: holomorphic approximation alone supplies no uniform bound. Finite total area is what allows the limiting map to extend across infinity; see Theorem 2.4.

Classical Brody reparametrization has a further limitation: it moves the source centers, so even disks through a fixed point need not yield an entire curve through that point. Duval addressed this loss of location by replacing derivative selection with control of where area accumulates [15]; see also [16]. His quantitative theorem localizes an entire curve on a compact set charged by an Ahlfors current. This is an important inspiration for our construction. In our finite-area setting, we retain position through a base constraint and two marked values, rather than invoke the Ahlfors-current theorem itself.

A second difficulty is to retain the prescribed geometric information in the limit. The analysis of bubbling goes back to Sacks and Uhlenbeck's work on harmonic maps [43] and Gromov's theory of pseudoholomorphic curves [24]. Bubbling can separate prescribed local data from nonconstant sphere components, while area may also concentrate near the source boundary. We must therefore obtain a nonconstant sphere satisfying both the required incidence conditions and the area bound.

Our construction addresses area control and localization together. We first use Ricci positivity to decrease area above an explicit threshold. The key is a Ricci-trace estimate, realized through approximation and interpolation by holomorphic families that preserve a scalar derivative condition and satisfy uniform estimates. To retain the geometric information under compactness, we use a relative formulation: the center may move in a compact fiber, while a derivative of the projection to the base remains normalized. This yields a sphere meeting but not contained in that fiber, with a sharp area bound; the pointwise specialization gives a rational curve through every prescribed point. The curves are constructed analytically over $\mathbb{C}$; the rational-connectedness and classification applications then use rational quotients and minimal families in characteristic zero. We explain the mechanism after stating the main results.

1.2. Main results

To prove rational connectedness by our method, we need more than a curve through a point: we need one that escapes a fiber of the rational quotient. Our main result therefore produces a holomorphic sphere meeting a compact fiber without being contained in it. We use the Ricci and area conventions in (1). The fiber in the statement is the full inverse image in the specified neighborhood.

Theorem 1.1. *Let (X, ω) be a compact Kähler n -fold satisfying*

$$\text{Ric}(\omega) \geq \kappa \omega, \quad \kappa > 0.$$

For an open subset $N \subset X$ and a complex manifold Y of positive dimension, consider a holomorphic map $q : N \rightarrow Y$ and a point $y_0 \in Y$. Suppose that the full fiber $F := q^{-1}(y_0)$ is nonempty and compact and that q is a submersion on a neighborhood of F in N . Then there is a nonconstant holomorphic map $u : \mathbb{P}^1 \rightarrow X$ such that

$$u(\mathbb{P}^1) \cap F \neq \emptyset, \quad u(\mathbb{P}^1) \not\subset F, \quad \int_{\mathbb{P}^1} u^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Yau's prescribed-Ricci theorem supplies a metric satisfying this hypothesis on every Fano manifold [48]; see Proposition 3.1. No Kähler–Einstein hypothesis is required. Conversely, the Ricci lower bound makes K_X^{-1} positive and hence ample, so the target is projective and the reduced image of the sphere is a rational curve; see Remark 3.2. The coefficient $2\pi(n+1)$ is sharp as a universal metric bound, as the projective-space example in Remark 2.8 shows. The relative viewpoint has a classical antecedent in Campana's generic-fiber escape criterion and his Fano theorem [6, Proposition 2.7 and Theorem 3.1]. Here we work near a specified compact full fiber of a holomorphic submersion and obtain a metric area bound.

Allowing the center to move within F lets the normalized base derivative pass to the principal limit whenever first-exit compactness has not already produced a sphere meeting but not contained in F . The final passage to the sharp area bound retains intersection with F and noncontainment in F , but need not preserve this derivative normalization. No tangent direction or contact order is prescribed.

Taking a coordinate map near a point gives the pointwise form immediately.

Theorem 1.2. *Let (X, ω) be a positive-dimensional compact Kähler manifold of complex dimension n , and suppose $\text{Ric}(\omega) \geq \kappa \omega$ for some $\kappa > 0$. Then, for every $x \in X$, there is a nonconstant*

holomorphic map $u_x : \mathbb{P}^1 \rightarrow X$ such that

$$x \in u_x(\mathbb{P}^1), \quad \int_{\mathbb{P}^1} u_x^* \omega \leq \frac{2\pi(n+1)}{\kappa}.$$

Classical bend-and-break produces a rational curve C through a prescribed point of a Fano n -fold with

$$0 < -K_X \cdot C \leq n+1.$$

Mori's account [40, p. 749] attributes this form of the result to Kollár, by a technique similar to that of [39]. Together with $\kappa\omega \leq \text{Ric}(\omega)$, this degree bound gives the same pointwise area estimate. The contribution here is an analytic construction of the curve with direct control of its area. See [11, Chapter 3] for the classical argument.

Starting from our analytic construction, we give proofs of the following two classical theorems in characteristic zero, without reduction to positive characteristic. Recall that a smooth projective variety is rationally connected if two general points lie on the image of one morphism $\mathbb{P}^1 \rightarrow X$. Fujiki formulated this notion for compact complex spaces [21, Definition 2.8 and the following remark]. The following theorem was proved by Campana and by Kollár, Miyaoka, and Mori [6, 35].

Theorem 1.3. *Every positive-dimensional complex Fano manifold is rationally connected.*

Campana's criterion [6, Proposition 2.7] reduces rational chain connectedness to the existence of a rational curve meeting but escaping a general fiber of every quasifibration with positive-dimensional base. Theorem 1.1 supplies precisely this curve, and characteristic-zero smoothing of rational chains gives rational connectedness.

The second application is Mori's theorem [39], which resolves Hartshorne's ample-tangent conjecture stated above.

Theorem 1.4. *Let X be a connected smooth complex projective variety of dimension $n \geq 1$. If T_X is ample, then $X \simeq \mathbb{P}^n$.*

If T_X is ample, then $K_X^{-1} = \det T_X$ is ample, so X is Fano. The pointwise construction supplies the first rational curve. Classical two-point degree reduction and Mori's minimal-curve classification then apply over $\mathbb{C}$; Section 6 checks their hypotheses and gives precise references.

1.3. From expanding disks to rational curves

The construction uses two disk surgeries. The first decreases area while preserving a scalar derivative normalization. The second uses the resulting margin below an area cap to enlarge the source and restore the normalization.

For $R > 0$, set $\mathbb{D}_R := \{z \in \mathbb{C} : |z| < R\}$. For a holomorphic map $f : \mathbb{D}_R \rightarrow X$, write $f_z := df(\partial/\partial z)$ and

$$\text{Area}_R(f) := \int_{\mathbb{D}_R} f^* \omega, \quad B_R(f) := \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_\omega.$$

The quantity B_R measures derivative concentration on the scale of the distance to the source boundary; it is the scale used in Brody's reparametrization argument [5].

For area descent, we first consider the variational problem with $f(0) = x$ and $\lambda(f_z(0)) = 1$ for a nonzero $\lambda \in T_x^* X$. The relative continuation uses the same pointwise calculation, but allows x to vary in F and takes λ to be the pullback of a fixed covector on the base. As in Brody's normalization, the nonzero scalar derivative prevents a smooth limit near the center from being constant; its transfer through bubbling is a separate compactness issue. In an adapted holomorphic frame with unitary boundary values, preserving the center requires every multiplier component to vanish at zero. Preserving one scalar derivative requires one component to vanish to second

order. Thus one quadratic and $n - 1$ linear components satisfy the constraints at total Dirichlet cost $\pi(n + 1)$. The corresponding paired second-variation trace is

$$4\pi(n + 1) - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \leq 4\pi(n + 1) - 2\kappa \text{Area}_R(f).$$

Thus the threshold

$$A_* := \frac{2\pi(n + 1)}{\kappa}$$

comes from the cost of scalar interpolation. The multiplier attains the sharp cost for its boundary and interpolation conditions (see the calculation after (17)); taking the trace makes positivity of the Ricci form sufficient. A negative infinitesimal trace alone does not provide the uniform area decrease needed for continuation. Theorem 4.8 realizes the calculation by actual holomorphic families whose paired second derivatives approximate the trace, with uniform parameter radius and cubic area remainder under fixed bounds on Area_R and B_R . A normalized projective lift, a corona theorem, and a holomorphic retraction provide this uniformity. This realization uses projectivity but does not require the Ricci inequality.

Fix an area cap $A > A_*$. For continuation, the center may move in the compact fiber F , and one derivative of the local projection to the base is normalized. Compactness gives a direct alternative: first-exit rescaling produces a sphere of area at most A meeting but not contained in F , or the normalization passes to the principal limit in the original source coordinate. At a finite limiting radius, area descent on this principal map leaves a margin below A . A first-jet perturbation of its affine lift then increases the scalar derivative, and source dilation restores its value to 1 on a larger disk. Continuity of total area keeps this second surgery below the cap. Thus a finite endpoint either yields the sphere or permits continuation. The precise relation between the two surgeries is explained after Lemma 2.2; successive disks need not extend one another.

On expanding sources, the same compactness alternative either yields the required sphere or gives a finite-area entire principal map. In the latter case, the retained base derivative makes the map nonconstant, and removal at infinity gives a sphere. The sharp bound requires a final limit $A \downarrow A_*$. At this stage we retain incidence rather than a derivative on the selected sphere. Two marked target values retain a rational chain from F to the boundary of a fixed neighborhood, with total area at most A_* . Its first component leaving F meets F and has area at most A_* . The compactness and removal theorems are supplied by Ivashkovich and Shevchishin [30]; their fixed-coordinate and marked consequences are derived in Section 5.

Figure 2 summarizes the two limiting stages for a point fiber.

Section 2 proves the relative sphere theorem and records the metric sharpness example. The Ricci-trace calculation, uniform holomorphic realization, and marked compactness are established in Sections 3–5. Section 6 gives the characteristic-zero proofs of rational connectedness and Hartshorne’s conjecture. Section 7 compares the analytic construction with Mori’s method and discusses the numerical positivity question.

2. PROOF OF THE RELATIVE SPHERE THEOREM

We prove Theorem 1.1 by continuing normalized holomorphic disks under a strict area bound. At each limiting stage, compactness either produces a sphere meeting but not contained in the fiber, or retains the scalar base derivative on a principal map to which the two surgeries apply. We state the disk operations and compactness results here and give their proofs in Sections 4 and 5.

2.1. Normalized disks and the two disk operations

We fix the metric normalization and recall the area and weighted derivative quantities from the Introduction.

Unless otherwise specified, (X, ω) is a compact Kähler manifold of complex dimension $n \geq 1$. In holomorphic coordinates $w^1, \dots, w^n$, let $h = (h_{\alpha\bar{\beta}})$ be the Gram matrix of the coordinate frame. Our

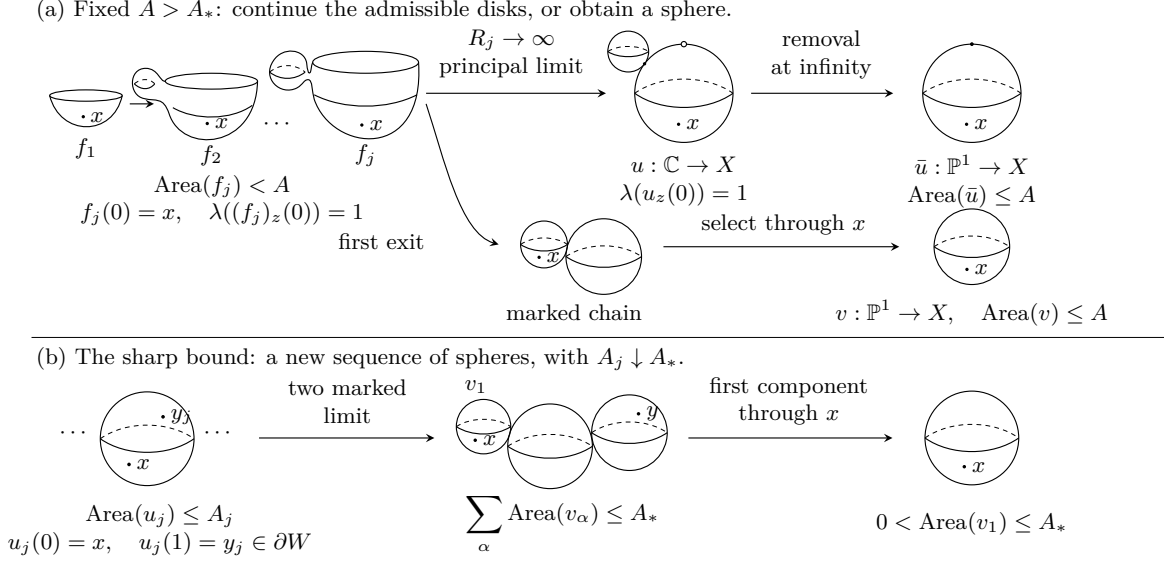


FIGURE 2. The two limiting stages, pictured for $F = \{x\}$. (a) Under a fixed cap, the compactness alternative supplies a marked chain and a sphere through x , or a normalized principal map. A finite normalized endpoint permits further continuation by the disk surgeries; on expanding sources, finite-area removal closes the entire principal map at infinity. The bulge and neck illustrate possible concentration; a sphere at the node represents a bubble obtained after reparametrization. (b) Two marked values retain a chain from x to $y \in \partial W$ as $A_j \downarrow A_*$, where W is a relatively compact neighborhood of x with smooth boundary inside a coordinate chart. Its first component has area at most A_* ; it need not contain y or retain the derivative normalization. All contours are schematic.

Hermitian inner products are conjugate-linear in the first variable and complex-linear in the second, so $h_{\alpha\bar{\beta}} = \langle \partial/\partial w^\alpha, \partial/\partial w^\beta \rangle_\omega$ and a coefficient column c has squared norm c^*hc . This convention accounts for the transposed indices in the formula for ω below. For a holomorphic source map f , write $f_z = df(\partial/\partial z)$. With repeated coordinate indices summed and dA denoting Euclidean area measure, our normalizations are

$$\omega = \frac{i}{2} h_{\beta\bar{\alpha}} dw^\alpha \wedge d\bar{w}^\beta, \quad f^*\omega = |f_z|_h^2 dA, \quad \text{Ric}(\omega) = -i\partial\bar{\partial} \log \det h. \quad (1)$$

For a holomorphic map $f : S \rightarrow X$ from a Riemann surface, its ω -area is $\text{Area}(f) := \int_S f^*\omega$, the geometric area counted with multiplicity. For $a \in \mathbb{C}$ and $r > 0$, write $\mathbb{D}_r(a) := \{z \in \mathbb{C} : |z - a| < r\}$, and abbreviate $\mathbb{D}_R := \mathbb{D}_R(0)$ and $\mathbb{D} := \mathbb{D}_1(0)$; bars denote closure, and $\partial\mathbb{D}_R$ is the boundary circle. We use the area on $\mathbb{D}_R$ and the boundary-distance-weighted derivative bound

$$\text{Area}_R(f) := \int_{\mathbb{D}_R} f^*\omega, \quad B_R(f) := \sup_{z \in \mathbb{D}_R} (R - |z|) |f_z(z)|_\omega. \quad (2)$$

Thus $B_R(f)$ weights the derivative by its distance to the source boundary. Write $\mathcal{O}(V, X)$ for the holomorphic maps from an open set $V \subset \mathbb{C}$ to X , and $K \Subset U$ when the closure of K is compact and contained in U . A holomorphic sphere is a map $\mathbb{P}^1 \rightarrow X$; a rational curve is the reduced image of a nonconstant such map. Smooth convergence means uniform convergence of all derivatives on compact source subsets.

Under the Ricci bound $\text{Ric}(\omega) \geq \kappa\omega$, $\kappa > 0$, set

$$A_* = \frac{2\pi(n+1)}{\kappa}. \quad (3)$$

Positive Ricci curvature makes the anticanonical bundle positive, and Kodaira's theorem makes X projective; see Remark 3.2. Fix a projective embedding for the two disk operations below.

For the relative constraint, we work in the proper setting obtained by the local reduction at the start of the proof of Theorem 1.1. Thus let $N \subset X$ be open and $q : N \rightarrow Y$ a proper holomorphic submersion to a complex manifold Y . Fix $y_0 \in q(N)$ and let $F = q^{-1}(y_0)$. After restricting Y to a coordinate ball about y_0 and N to its inverse image, write $d_Y = \dim Y$ and let $\psi : N \rightarrow \mathbb{C}^{d_Y}$ be the coordinate expression of q , translated so that $\psi^{-1}(0) = F$. Fix $0 \neq \eta \in (\mathbb{C}^{d_Y})^*$ and set

$$\lambda^F = (\lambda_x^F)_{x \in F}, \quad \lambda_x^F := \eta \circ d\psi_x \in T_x^*X.$$

The center $f(0)$ may vary in F , but the base covector η remains fixed. Near zero, the condition $\lambda_{f(0)}^F(f_z(0)) = 1$ is precisely $(\eta \circ \psi \circ f)_z(0) = 1$. In particular, $d\psi_{f(0)}(f_z(0)) \neq 0$, so the initial tangent vector is not tangent to F . The submersion property makes λ_x^F nonzero for every $x \in F$. The classes below consist of maps on $\mathbb{D}_R$ that extend holomorphically to some open neighborhood of $\overline{\mathbb{D}}_R$. The neighborhood may depend on the map. Set

$$\mathcal{A}_R^{F, \lambda^F} := \left\{ f \in \mathcal{O}(\mathbb{D}_R, X) : f \text{ extends holomorphically near } \overline{\mathbb{D}}_R, \right. \\ \left. f(0) \in F, \quad \lambda_{f(0)}^F(f_z(0)) = 1 \right\}, \quad (4)$$

$$a_{F, \lambda^F}(R) := \inf \{ \text{Area}_R(f) : f \in \mathcal{A}_R^{F, \lambda^F} \}.$$

We use the convention $\inf \emptyset = +\infty$. Restricting a competitor from $\mathbb{D}_t$ to $\mathbb{D}_s$, with $0 < s < t$, preserves admissibility and cannot increase its area. Taking infima gives $a_{F, \lambda^F}(s) \leq a_{F, \lambda^F}(t)$. Thus the area profile is nondecreasing, with left limit at each finite $R > 0$ given by

$$a_{F, \lambda^F}(R^-) := \sup_{0 < s < R} a_{F, \lambda^F}(s).$$

We use almost minimizers without assuming that the infimum is attained: an ε -almost minimizer at radius R is a map in $\mathcal{A}_R^{F, \lambda^F}$ with area less than $a_{F, \lambda^F}(R) + \varepsilon$. To start the construction, choose a local coordinate disk satisfying $\lambda_{f(0)}^F(f_z(0)) = 1$. It is admissible on some disk of positive radius, and restriction to smaller disks gives

$$\lim_{R \downarrow 0} a_{F, \lambda^F}(R) = 0. \quad (5)$$

Continuation compares admissible disks at different radii; the disks need not extend or approximate a common initial map. Area descent above A_* creates a strict margin, and scalar enlargement uses that margin to increase the source radius.

The first operation lowers area by a uniform positive amount whenever the area exceeds A_* by a fixed amount and both area and weighted derivative have fixed upper bounds.

Proposition 2.1. *Assume $\text{Ric}(\omega) \geq \kappa\omega$ with $\kappa > 0$ and keep the fixed projective embedding of X . For every $A_0, B_0, \delta > 0$, there is $\sigma > 0$ with the following property: for every $R > 0$, each $f \in \mathcal{A}_R^{F, \lambda^F}$ satisfying*

$$A_* + \delta \leq \text{Area}_R(f) \leq A_0, \quad B_R(f) \leq B_0,$$

admits a competitor $g \in \mathcal{A}_R^{F, \lambda^F}$ with $\text{Area}_R(g) \leq \text{Area}_R(f) - \sigma$.

The second operation starts with one finite-area map on an open disk. A holomorphic perturbation fixes its center and increases its scalar derivative. Source dilation then restores the normalization and enlarges the disk. Only continuity of the total area of this perturbation is needed.

Lemma 2.2. *Keep the projective embedding of X and the relative constraint above. Let $0 < R < \infty$ and let $h : \mathbb{D}_R \rightarrow X$ be holomorphic, with*

$$\text{Area}_R(h) < \infty, \quad h(0) \in F, \quad \lambda_{h(0)}^F(h_z(0)) = 1.$$

There are $t_0 > 0$ and holomorphic maps $h_t : \mathbb{D}_R \rightarrow X$, $0 \leq t < t_0$, such that $h_0 = h$ and

$$h_t(0) = h(0), \quad \lambda_{h_t(0)}^F((h_t)_z(0)) = 1 + t, \quad \lim_{t \downarrow 0} \text{Area}_R(h_t) = \text{Area}_R(h).$$

Consequently,

$$G_t(z) := h_t(z/(1+t)), \quad |z| < (1+t)R,$$

has center in F , scalar derivative 1, and total area $\text{Area}_R(h_t)$. For every $A > \text{Area}_R(h)$, some $R' > R$ admits a map $g \in \mathcal{A}_{R'}^{F, \lambda^F}$ with $\text{Area}_{R'}(g) < A$.

The maps h and h_t need not extend past $\partial\mathbb{D}_R$. For the last assertion, take $t > 0$ with $\text{Area}_R(h_t) < A$ and restrict G_t to any R' strictly between R and $(1+t)R$. This restriction extends holomorphically past its closed source disk. The proof in Section 4.3 constructs the perturbation by adding $tzH'(0)$ to an affine lift H of h .

The two operations require different forms of area control. When compactness retains a normalized principal limit h of almost minimizers at a finite endpoint R , the argument below proves

$$\text{Area}_R(h) = L := a_{F, \lambda^F}(R^-), \quad B_R(h) < \infty,$$

the second assertion being Lemma 2.7. Restrictions of this one map share the bounds L and $B_R(h)$. If $L > A_*$, Proposition 2.1 supplies a fixed area drop on restrictions close to R , contradicting the limiting infimum L . Thus $L \leq A_*$. For a cap $A > A_*$, the remaining margin permits Lemma 2.2 to enlarge this individual disk by continuity of total area. No common bound for the weighted derivatives of the original almost minimizers is needed. The enlargement must still respect the cap: an uncontrolled area increase would not permit continuation.

Figure 3 separates the holomorphic perturbation from the source dilation.

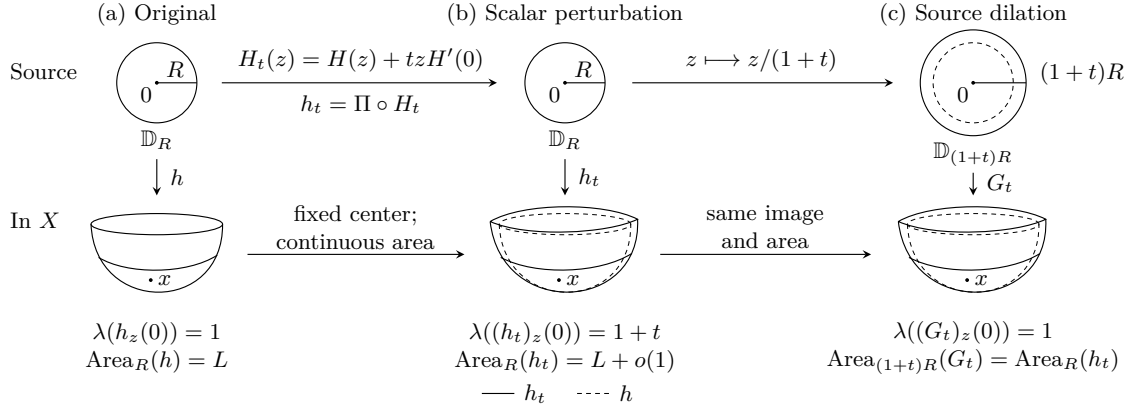


FIGURE 3. Enlargement of one finite-area disk at a fixed center $x \in F$, with $\lambda = \lambda_x^F$. The affine-lift perturbation $h_t = \Pi(H + tzH'(0))$ acts on the same disk and increases the scalar derivative from 1 to $1 + t$. Dilation gives $G_t(z) = h_t(z/(1+t))$ on $\mathbb{D}_{(1+t)R}$, with scalar derivative 1 and unchanged total area. Here $o(1) \rightarrow 0$ as $t \downarrow 0$ for this fixed h . The outlines are schematic; dashed target outlines show h .

2.2. Principal limits and marked values

Compactness serves two purposes here: it produces a principal limit in the original source coordinate, and it preserves a connection between two marked target sets when bubbling occurs. Finite-area removal then turns an entire principal map into a holomorphic sphere. The three statements require only a compact Kähler target and are proved in Section 5.

Proposition 2.3. *Let (X, ω) be compact Kähler. For each j , let $R_j > 0$ and let $f_j : U_j \rightarrow X$ be holomorphic, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$. Assume that $\sup_j \text{Area}_{R_j}(f_j) < \infty$, and set $A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Choose $R_\infty \in (0, \infty]$ with $R_\infty \leq \liminf_j R_j$, interpreting $\mathbb{D}_\infty$ as $\mathbb{C}$. After passing to a subsequence realizing A_∞ and then to a further subsequence, there are a finite set $\Sigma_\infty \subset \mathbb{D}_{R_\infty}$ and a holomorphic map $f_\infty : \mathbb{D}_{R_\infty} \rightarrow X$ such that, in the original source coordinate,*

$$f_j \longrightarrow f_\infty \quad \text{smoothly on compact subsets of } \mathbb{D}_{R_\infty} \setminus \Sigma_\infty, \quad \text{Area}_{R_\infty}(f_\infty) \leq A_\infty.$$

The principal limit f_∞ is holomorphic on all of $\mathbb{D}_{R_\infty}$ and may be constant. The asserted convergence of f_j holds only away from Σ_∞ ; convergence at its points is not claimed.

Theorem 2.4. *Let X be compact Kähler. A holomorphic map from a punctured disk to X with finite area extends holomorphically over the puncture. Consequently, a finite-area holomorphic map $u : \mathbb{C} \rightarrow X$ extends to a holomorphic map $\mathbb{P}^1 \rightarrow X$.*

Lemma 2.5. *Let K_0, K_1 be disjoint compact subsets of a compact Kähler manifold X . Suppose $R_j > 1$, $R_j \rightarrow \infty$, and the holomorphic maps $f_j : U_j \rightarrow X$, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$, satisfy*

$$f_j(0) \in K_0, \quad f_j(1) \in K_1, \quad \sup_j \text{Area}_{R_j}(f_j) < \infty.$$

Then there are nonconstant holomorphic maps $u_1, \dots, u_\ell : \mathbb{P}^1 \rightarrow X$ whose images form a chain: the first meets K_0 , the last meets K_1 , and consecutive images intersect. Their total area, counted with the multiplicities of these maps, satisfies

$$\sum_{i=1}^{\ell} \text{Area}(u_i) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

The fixed source marks and the disjoint target sets play different roles. Since the marks remain at 0 and 1, stabilization cannot produce an attached tree made entirely of constant components: such a stable tree would need at least two marks converging to the same source point. Since K_0 and K_1 are disjoint, the nodal path between the marked values must contain a nonconstant component.

The scalar derivative is a derivative of the bounded local base map. This gives a direct alternative: a first exit from a neighborhood of F produces a sphere, while trapping near F allows the scalar constraint to pass to the principal map even if the original maps bubble at zero.

Lemma 2.6. *Let (X, ω) be compact Kähler and retain F , N , ψ , and λ^F from (4). Suppose $R_j \rightarrow R_\infty \in (0, \infty]$ and $f_j \in \mathcal{A}_{R_j}^{F, \lambda^F}$, with*

$$\text{Area}_{R_j}(f_j) \longrightarrow E < \infty.$$

Then at least one of the following conclusions holds:

- (i) *There is a nonconstant sphere $v : \mathbb{P}^1 \rightarrow X$ such that*

$$v(\mathbb{P}^1) \cap F \neq \emptyset, \quad v(\mathbb{P}^1) \not\subset F, \quad \text{Area}(v) \leq E.$$

- (ii) *A subsequence has a principal limit $h : \mathbb{D}_{R_\infty} \rightarrow X$ in the original source coordinate, as in Proposition 2.3, satisfying*

$$\text{Area}_{R_\infty}(h) \leq E, \quad h(0) \in F, \quad \lambda_{h(0)}^F(h_z(0)) = 1.$$

On some fixed disk about zero, the base compositions $\psi \circ f_j$ converge to $\psi \circ h$ smoothly on compact subsets. Here $\mathbb{D}_\infty = \mathbb{C}$.

Proof. Choose $F \Subset V \Subset N$ with smooth boundary, and fix $r > 0$ such that $R_j > r$ for all sufficiently large j . There are two possibilities after passage to a subsequence.

First, suppose there are $z_j \rightarrow 0$ with $z_j \neq 0$ and $f_j(z_j) \in \partial V$. Such points can be chosen as first exits along segments from zero whenever no fixed small closed source disk has image in V along an infinite subsequence. The rescaled maps

$$v_j(\zeta) := f_j(z_j \zeta), \quad |\zeta| < R_j/|z_j|,$$

extend past their closed source disks and satisfy

$$v_j(0) \in F, \quad v_j(1) \in \partial V, \quad \text{Area}_{R_j/|z_j|}(v_j) = \text{Area}_{R_j}(f_j) \rightarrow E.$$

Their source radii tend to infinity. Lemma 2.5 gives a chain of nonconstant spheres from F to ∂V , of total area at most E . Take its first component not contained in F . It meets F , either at the first marked value or at its intersection with the preceding component, and has area at most E . This proves (i).

Otherwise, for some $0 < \rho < r$, an infinite subsequence satisfies $f_j(\overline{\mathbb{D}}_\rho) \subset V$. Apply Proposition 2.3 on $\mathbb{D}_{R_\infty}$ to this subsequence. Its principal map h takes values in $\overline{V}$ on $\mathbb{D}_\rho$: this holds away from the finite concentration set by convergence, and at that set by continuity. In particular, $\psi \circ h$ is defined there. The bounded holomorphic maps $\psi \circ f_j$ have a further locally uniformly convergent subsequence. Principal convergence identifies its limit with $\psi \circ h$ away from the concentration set; the identity theorem identifies them throughout $\mathbb{D}_\rho$. Cauchy estimates give smooth convergence on compact subdisks. Passing the value and scalar derivative at zero to the limit yields

$$\psi(h(0)) = 0, \quad (\eta \circ \psi \circ h)_z(0) = \lim_j (\eta \circ \psi \circ f_j)_z(0) = 1.$$

The area bound follows from principal compactness, proving (ii). $\square$

The following lemma supplies the weighted derivative bound for the principal map at a finite endpoint.

Lemma 2.7. *Let (X, ω) be compact Kähler, let $0 < R < \infty$, and let $h : \mathbb{D}_R \rightarrow X$ be holomorphic with $\text{Area}_R(h) < \infty$. Then $B_R(h) < \infty$.*

The proof in Section 5.4 uses point selection: an unbounded weighted derivative for this fixed map would place a positive amount of area in arbitrarily thin boundary collars.

2.3. Continuation and the sharp area bound

Proof of Theorem 1.1. Keep the threshold A_* and the fixed projective embedding chosen at the beginning of this section. Choose $F \Subset U \Subset N$ with smooth boundary, keeping $\overline{U}$ inside the submersion neighborhood. The full-fiber hypothesis gives $y_0 \notin q(\partial U)$. We can therefore choose a coordinate ball Y' about y_0 such that $\overline{Y'} \cap q(\partial U) = \emptyset$ and $Y' \subset q(U)$. After replacing N by $U \cap q^{-1}(Y')$ and Y by Y' , the map q is proper: the inverse image of a compact subset of Y' is closed in $\overline{U}$ and avoids ∂U . In these restricted neighborhoods, take the coordinates $\psi : N \rightarrow \mathbb{C}^{d_Y}$ and a nonzero covector $\eta \in (\mathbb{C}^{d_Y})^*$, and use the classes (4).

Continuation under a strict cap.

Fix $A > A_*$ and abbreviate $a(R) = a_{F, \lambda^F}(R)$. Set

$$\mathcal{S}_A := \{R > 0 : a(R) < A\}, \quad R_A := \sup \mathcal{S}_A.$$

Small coordinate disks give $\mathcal{S}_A \neq \emptyset$, and restriction supplies every smaller positive radius. Each $R \in \mathcal{S}_A$ also admits a slightly larger radius: choose $f \in \mathcal{A}_R^{F, \lambda^F}$ with $\text{Area}_R(f) < A$, and extend the source slightly within the holomorphic neighborhood of $\mathbb{D}_R$. Continuity of its area preserves the strict cap. Thus

$$\mathcal{S}_A = (0, R_A). \tag{6}$$

Consider first $R_A < \infty$. Choose $r_j \uparrow R_A$ and write

$$L := a(R_A^-) = \sup_{0 < s < R_A} a(s) \leq A.$$

Choose $0 < \varepsilon_j < \min\{j^{-1}, (A - a(r_j))/2\}$ and $f_j \in \mathcal{A}_{r_j}^{F, \lambda^F}$ such that

$$a(r_j) \leq \text{Area}_{r_j}(f_j) < a(r_j) + \varepsilon_j < A.$$

Then $\text{Area}_{r_j}(f_j) \rightarrow L$. Apply Lemma 2.6. Its first outcome already gives a sphere meeting but not contained in F , with area at most $L \leq A$. We show that the second outcome permits continuation beyond R_A .

The finite endpoint.

In the second outcome, let $h : \mathbb{D}_{R_A} \rightarrow X$ be the normalized principal map. For every $0 < s < R_A$, its restriction extends holomorphically past $\overline{\mathbb{D}}_s$ and belongs to $\mathcal{A}_s^{F, \lambda^F}$. Hence

$$a(s) \leq \text{Area}_s(h), \quad \text{Area}_{R_A}(h) \leq L.$$

Letting $s \uparrow R_A$ gives $\text{Area}_{R_A}(h) = L$. Together with Lemma 2.7, this yields precisely the endpoint data needed below:

$$\text{Area}_{R_A}(h) = L, \quad B_{R_A}(h) < \infty. \quad (7)$$

If $L > A_*$, set $\delta = (L - A_*)/2$. For $s < R_A$ sufficiently close to R_A , we have

$$A_* + \delta \leq \text{Area}_s(h) \leq A, \quad B_s(h) \leq B_{R_A}(h).$$

Proposition 2.1, with these common caps, gives a fixed $\sigma > 0$ and competitors $g_s \in \mathcal{A}_s^{F, \lambda^F}$ satisfying

$$a(s) \leq \text{Area}_s(g_s) \leq \text{Area}_s(h) - \sigma \leq L - \sigma.$$

Since $a(s) \rightarrow L$, this is impossible. Therefore

$$L \leq A_* < A. \quad (8)$$

Apply Lemma 2.2 to h . The strict margin in (8) and continuity of $\text{Area}_{R_A}(h_t)$ give a small $t > 0$ with $\text{Area}_{R_A}(h_t) < A$. The normalized map

$$G_t(z) = h_t(z/(1+t))$$

is holomorphic on $\mathbb{D}_{(1+t)R_A}$ and has this same total area. Choose $R_A < R' < (1+t)R_A$ and restrict G_t to $\mathbb{D}_{R'}$. The resulting competitor belongs to $\mathcal{A}_{R'}^{F, \lambda^F}$ and has area less than A , contrary to the definition of R_A . Thus at a finite endpoint the alternative supplies the required sphere.

Expanding source disks.

If $R_A = \infty$, choose $R_j \rightarrow \infty$ and $f_j \in \mathcal{A}_{R_j}^{F, \lambda^F}$ with $\text{Area}_{R_j}(f_j) < A$. Pass to a subsequence whose areas converge to $E \leq A$, and apply Lemma 2.6. The first outcome again supplies the required sphere. In the second, the principal map $u : \mathbb{C} \rightarrow X$ satisfies

$$\text{Area}(u) \leq E, \quad u(0) \in F, \quad (\eta \circ \psi \circ u)_z(0) = \lambda_{u(0)}^F(u_z(0)) = 1,$$

where the base composition is used only near zero. Theorem 2.4 extends u across infinity. Its nonzero base derivative shows that the resulting sphere meets F without being contained in F . We have therefore obtained such a sphere of area at most A for every $A > A_*$.

The sharp area bound.

To reach the exact bound A_* , we take a second limit, now retaining incidence through two marked values rather than requiring the scalar derivative normalization to survive. For each j , choose a sphere $u_j : \mathbb{P}^1 \rightarrow X$ meeting F without being contained in F , with

$$\int_{\mathbb{P}^1} u_j^* \omega \leq A_* + \frac{1}{j}.$$

Choose $F \Subset W_F \Subset N$ with smooth boundary. A sphere meeting F without being contained in F must leave W_F , and hence meet ∂W_F . Indeed, if $u_j(\mathbb{P}^1) \subset W_F$, the holomorphic map $\psi \circ u_j : \mathbb{P}^1 \rightarrow \mathbb{C}^{d_Y}$ is constant. Since it vanishes at a point mapped into F , it vanishes everywhere, forcing $u_j(\mathbb{P}^1) \subset F$, a contradiction.

To put the spheres in the form needed for two-marked compactness, choose distinct points $p_j, q_j \in \mathbb{P}^1$ mapped into F and ∂W_F , respectively, and reparametrize so that these source marks are 0 and 1. Let $f_j : \mathbb{C} \rightarrow X$ be the restriction of the reparametrized sphere to the affine chart. Taking $R_j = j + 1$ then gives

$$f_j(0) \in F, \quad f_j(1) \in \partial W_F, \quad \text{Area}_{R_j}(f_j) \leq A_* + \frac{1}{j}.$$

Applying Lemma 2.5 to these fixed marks with $K_0 = F$ and $K_1 = \partial W_F$ gives a chain $v_1, \dots, v_\ell$ such that

$$\sum_{\alpha=1}^{\ell} \int_{\mathbb{P}^1} v_\alpha^* \omega \leq A_*.$$

Follow the chain from F toward ∂W_F and take the first component not contained in F :

$$k := \min\{\alpha : v_\alpha(\mathbb{P}^1) \not\subset F\}.$$

Such a component exists because the last one meets ∂W_F . It still meets F : if it is the first component, this follows from the marked value; otherwise, it shares a nodal value with its predecessor, whose image lies in F . Hence

$$v_k(\mathbb{P}^1) \cap F \neq \emptyset, \quad v_k(\mathbb{P}^1) \not\subset F, \quad 0 < \text{Area}(v_k) \leq \sum_{\alpha=1}^{\ell} \text{Area}(v_\alpha) \leq A_*.$$

Thus v_k meets F without being contained in F and has area at most A_* , as required. The two marked values are retained by the chain; the selected component need not contain both of them or retain the original derivative normalization. $\square$

Proof of Theorem 1.2. Given $x \in X$, let N be a coordinate neighborhood of x and let $q : N \rightarrow Y \subset \mathbb{C}^n$ be its coordinate map. The full fiber over $q(x)$ is $F = \{x\}$, and q is a submersion. Theorem 1.1 therefore gives a nonconstant sphere through x with area at most A_* . $\square$

Remark 2.8. On $\mathbb{P}^n$, use the Fubini–Study form

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log(1 + |w^1|^2 + \dots + |w^n|^2).$$

In the normalization (1), $\text{Ric}(\omega_{\text{FS}}) = 2(n+1)\omega_{\text{FS}}$ and $\int_{\mathbb{P}^1} \omega_{\text{FS}} = \pi$ on a projective line. Every nonconstant sphere in $\mathbb{P}^n$ has positive integer degree d and area $d\pi$. Thus for $\kappa = 2(n+1)$ the bound $2\pi(n+1)/\kappa = \pi$ is attained by a line and cannot be decreased.

Remark 2.9. The finite-radius bound in (7) does not assert a common weighted derivative bound on expanding normalized disks. Indeed, if $M_{\lambda, F} = \max_{x \in F} \|\lambda_x^F\|_{\omega, *}$, then normalization gives

$$B_R(f) \geq R|f_z(0)|_\omega \geq R/M_{\lambda, F} \quad (f \in \mathcal{A}_R^{F, \lambda^F}).$$

The expanding-source argument instead retains the local base derivative on the principal map.

Remark 2.10. For fixed $A > A_*$, one may assume that every sphere of area at most A meeting F is contained in F . This excludes the first outcome of Lemma 2.6. The same descent and enlargement then force $R_A = \infty$, and removal of the normalized entire principal map contradicts that assumption. The passage $A \downarrow A_*$ is unchanged.

The direct proof allows each compactness step to produce the sphere and makes no initial nonexistence assumption. It still uses infima, subsequences, and incompatible inequalities at a putative finite endpoint; it does not provide a finite algorithm for producing the sphere.

3. RICCI TRACES AND SCALAR-CONSTRAINED VARIATIONS

The Ricci trace separates curvature from the cost of fixing a value and one scalar derivative. An adapted multiplier has Dirichlet cost $\pi(n+1)$; Proposition 3.4 turns this into the descent estimate. Section 4 realizes the variation fields by holomorphic families. We retain the normalizations (1)–(2).

For $z = x + iy$, the complex derivative and the Euclidean Laplacian are

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \quad \Delta = \partial_x^2 + \partial_y^2 = 4\partial_z\partial_{\bar{z}}.$$

We write T_X for the holomorphic tangent bundle, $T_x X$ for its fiber at x , and T_X^* for its dual. The pullback f^*T_X has fiber $T_{f(z)}X$ and carries the Chern connection induced by ω ; ∇_z denotes covariant differentiation in the source direction. On $\partial\mathbb{D}_R$, ν is the outward Euclidean unit normal, ∂_ν the corresponding derivative, and ds Euclidean arclength. Let A^* denote the conjugate transpose of A . We use the Hilbert–Schmidt norm and the operator norm

$$\|A\|_{\text{HS}}^2 := \text{tr}(AA^*) = \sum_{j,k} |A_{jk}|^2, \quad \|A\|_{\text{op}} := \sup_{|v|=1} |Av|. \quad (9)$$

The symbol Id_n denotes the identity matrix. Distances and diameters use the specified metric.

A holomorphic disk is a holomorphic map from a disk in $\mathbb{C}$ to X . An entire curve is a nonconstant holomorphic map $\mathbb{C} \rightarrow X$.

We write $\mathbb{P}^k$ for the space of complex lines in $\mathbb{C}^{k+1}$, with homogeneous coordinates $[Z_0 : \cdots : Z_k]$. The canonical bundle is $K_X = \bigwedge^n T_X^*$; its dual K_X^{-1} is also written $-K_X$.

For a Hermitian holomorphic line bundle (L, h) , we use the real Chern curvature form $-i\partial\bar{\partial}\log|s|_h^2$, where s is a nonvanishing local holomorphic frame. Its cohomology class is $2\pi c_1(L)$. We write $c_1(X) = c_1(K_X^{-1})$.

Yau’s theorem supplies the Ricci lower bound that allows the relative sphere theorem to be applied to Fano manifolds.

Proposition 3.1. *Every complex Fano manifold admits a Kähler metric ω and a constant $\kappa > 0$ such that*

$$\text{Ric}(\omega) \geq \kappa\omega. \quad (10)$$

Proof. Ampleness of K_X^{-1} gives a positive closed $(1,1)$ -form ρ representing $2\pi c_1(X)$. In a fixed Kähler class, Yau’s prescribed-Ricci theorem [48] gives a metric ω with $\text{Ric}(\omega) = \rho$. The least eigenvalue of ρ relative to ω is positive and continuous on the compact manifold. Its positive minimum therefore supplies κ with $\text{Ric}(\omega) \geq \kappa\omega$. $\square$

Remark 3.2. A connected compact Kähler manifold with a metric satisfying (10) is Fano. The induced Hermitian metric on K_X^{-1} has positive Chern curvature $\text{Ric}(\omega)$, so Kodaira’s embedding theorem makes K_X^{-1} ample and X projective [33, pp. 314–316].

For a holomorphic family on a fixed source disk, the Kähler condition expresses the paired second area variation as a boundary integral. The boundary values are free, and no critical-point assumption is made. This identity lets the trace calculation use variation fields before constructing their holomorphic families.

Lemma 3.3. *Fix $R > 0$. Let $\Delta_t \subset \mathbb{C}$ be a parameter disk centered at zero, $U \subset \mathbb{C}$ an open neighborhood of $\mathbb{D}_R$, and $\mathcal{F} : U \times \Delta_t \rightarrow X$ a holomorphic map. Write $f_t(z) := \mathcal{F}(z, t)$ and $\xi := \partial_t \mathcal{F}(\cdot, 0)$. Denote the second derivatives along the two real parameter axes by*

$$H_{\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_s), \quad H_{i\mathbb{R}} := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_{is}).$$

Then

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = \int_{\partial\mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (11)$$

For a fixed central map f_0 , the sum $H_{\mathbb{R}} + H_{i\mathbb{R}}$ therefore depends only on the variation field ξ , not on the second-order parameter data of the family.

Proof. Set $\tilde{\omega} := \mathcal{F}^*\omega$, and write its Hermitian coefficient matrix in the source coordinates $\zeta^1 = z$, $\zeta^2 = t$ as

$$\tilde{\omega} = \frac{i}{2} \sum_{\alpha, \beta=1}^2 \tilde{\omega}_{\alpha\bar{\beta}} d\zeta^\alpha \wedge d\bar{\zeta}^\beta.$$

Since $d\tilde{\omega} = 0$, comparison of its $(2, 1)$ and $(1, 2)$ coefficients gives

$$\partial_t \tilde{\omega}_{z\bar{z}} = \partial_z \tilde{\omega}_{t\bar{z}}, \quad \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_{\bar{z}} \tilde{\omega}_{t\bar{t}}.$$

Differentiating and substituting gives

$$\partial_t \partial_{\bar{t}} \tilde{\omega}_{z\bar{z}} = \partial_z \partial_{\bar{t}} \tilde{\omega}_{t\bar{z}} = \partial_z \partial_{\bar{z}} \tilde{\omega}_{t\bar{t}}.$$

At $t = 0$, $\tilde{\omega}_{t\bar{t}} = |\xi|_\omega^2$. Consequently

$$\partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0} = \int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} |\xi|_\omega^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds.$$

The sum of the second derivatives along the real and imaginary parameter axes is the parameter Laplacian $4\partial_t \partial_{\bar{t}}$. Hence

$$H_{\mathbb{R}} + H_{i\mathbb{R}} = 4 \partial_t \partial_{\bar{t}} \text{Area}_R(f_t)|_{t=0},$$

which yields (11). $\square$

Fix $R > 0$ and a map f holomorphic on a neighborhood of $\overline{\mathbb{D}_R}$. Without choosing a deformation family, we can define the following boundary quadratic form on holomorphic sections ξ of f^*T_X that are smooth on the closed disk:

$$\mathcal{H}_{R,f}(\xi) := \int_{\partial \mathbb{D}_R} \partial_\nu |\xi|_\omega^2 ds. \quad (12)$$

When ξ is the tangent field of a complex holomorphic family, Lemma 3.3 identifies $\mathcal{H}_{R,f}(\xi)$ with the sum of the second area derivatives along the real and imaginary parameter axes.

A boundary-unitary holomorphic frame, supplied by Lemma 4.1, makes the curvature term a Ricci integral after summation. The multiplier contributes only its Dirichlet cost, so a suitable interpolation multiplier will turn the identity into the descent threshold.

Proposition 3.4. *Let $e = (e_1, \dots, e_n)$ be a holomorphic frame of f^*T_X , smooth on the closed disk and unitary on $\partial \mathbb{D}_R$. For a positive integer N , let Φ be a holomorphic $\text{Mat}_{n \times N}(\mathbb{C})$ -valued map on a neighborhood of $\overline{\mathbb{D}_R}$, satisfying the boundary coisometry condition*

$$\Phi(\zeta) \Phi(\zeta)^* = \text{Id}_n \quad (|\zeta| = R).$$

The columns Φ_a give sections $(e\Phi)_a = \sum_{j=1}^n e_j \Phi_{ja}$ satisfying the trace identity

$$\sum_{a=1}^N \mathcal{H}_{R,f}((e\Phi)_a) = 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega). \quad (13)$$

Here Φ' is the entrywise derivative and $\|\cdot\|_{\text{HS}}$ is the norm in (9).

Proof. With our convention that the metric is linear in its second variable, the Gram matrix $G := (\langle e_j, e_k \rangle)_{j,k}$ represents squared norms by $|ec|_\omega^2 = c^* G c$. Summing (12) over the columns gives the boundary integral of $\partial_\nu \text{tr}(G \Phi \Phi^*)$. On the boundary, $G = \Phi \Phi^* = \text{Id}_n$, but their normal derivatives need not vanish. The product rule and Jacobi's determinant identity therefore give, on $\partial \mathbb{D}_R$,

$$\begin{aligned} \partial_\nu \text{tr}(G \Phi \Phi^*) &= \text{tr}(\partial_\nu G) + \text{tr}(\partial_\nu (\Phi \Phi^*)), \\ \partial_\nu \log \det G &= \text{tr}(G^{-1} \partial_\nu G) = \text{tr}(\partial_\nu G). \end{aligned}$$

The divergence theorem and $\Delta \operatorname{tr}(\Phi \Phi^*) = 4 \|\Phi'\|_{\text{HS}}^2$ give the multiplier term. Since $\det G$ is the metric coefficient on $f^* K_X^{-1}$, (1) gives

$$f^* \operatorname{Ric}(\omega) = -\frac{1}{2} \Delta \log \det G \, dA.$$

Green's formula therefore turns the determinant term $\partial_\nu \log \det G$ into $-2 \int f^* \operatorname{Ric}(\omega)$. $\square$

We first construct variations that fix the value of a disk at 0 and one nonzero scalar component of its derivative. Fix $x_0 \in X$ and a nonzero complex-linear covector

$$\lambda \in T_{x_0}^* X \setminus \{0\}. \quad (14)$$

The scalar condition $\lambda(f_z(0)) = 1$ fixes one component of the initial derivative and gives $|f_z(0)|_\omega \geq \|\lambda\|_{\omega, *}^{-1}$, where the norm on the covector is dual to ω at x_0 . The multiplier below imposes the linearized value and scalar-derivative constraints. Theorem 4.9 realizes these constraints by holomorphic families that preserve $f(0) = x_0$ and $\lambda(f_z(0)) = 1$ exactly, while leaving the other derivative components free. In Section 2, the center varies in a compact fiber and the covector is pulled back from the base. In the principal branch of Lemma 2.6, this base normalization passes to the limit. For a disk satisfying $f(0) = x_0$ and $\lambda(f_z(0)) = 1$, the frame value $e(0) : \mathbb{C}^n \rightarrow T_{x_0} X$ identifies the scalar constraint with the nonzero covector $L_f := \lambda \circ e(0)$. Choose $U_f \in U(n)$ so that its last $n-1$ columns form an orthonormal basis of $\ker L_f$. This leaves one direction subject to the scalar constraint. In this adapted basis, set

$$\Phi_f(z) := U_f \operatorname{diag} \left((z/R)^2, z/R, \dots, z/R \right). \quad (15)$$

The columns vanish at zero, so $(e\Phi_f)'(0) = e(0)\Phi_f'(0)$. The first column has zero derivative; the others have derivatives in $\ker L_f$. Thus $\Phi_f(0) = 0$ and $L_f \Phi_f'(0) = 0$ impose the infinitesimal constraints. On $|z| = R$, the diagonal entries have modulus one, so $\Phi_f \Phi_f^* = \operatorname{Id}_n$. For each integer $d \geq 1$, scaling $z = R\zeta$ and integrating in polar coordinates gives $\int_{\mathbb{D}_R} |((z/R)^d)'|^2 dA = \pi d$, independently of R . Since U_f is unitary, the quadratic column contributes 2π and each of the $n-1$ linear columns contributes π . Hence

$$\int_{\mathbb{D}_R} \|\Phi_f'\|_{\text{HS}}^2 dA = 2\pi + (n-1)\pi = \pi(n+1). \quad (16)$$

If $\operatorname{Ric}(\omega) \geq \kappa\omega$, then

$$\sum_{a=1}^n \mathcal{H}_{R,f}((e\Phi_f)_a) \leq 4\pi(n+1) - 2\kappa \operatorname{Area}_R(f). \quad (17)$$

The cost in (16) is optimal within the prescribed multiplier class. Indeed, fix $R > 0$, $N \geq 1$, and $0 \neq L \in (\mathbb{C}^N)^*$, and let Φ be holomorphic near $\mathbb{D}_R$ with values in $\operatorname{Mat}_{n \times N}(\mathbb{C})$, satisfying

$$\Phi \Phi^* = \operatorname{Id}_n \quad \text{on } \partial \mathbb{D}_R, \quad \Phi(0) = 0, \quad L\Phi'(0) = 0.$$

The boundary identity forces $N \geq n$. Choose $Q \in U(n)$ with first row $L/\|L\|$, where $\|L\|$ is the Euclidean dual norm. Multiplication by Q preserves that identity and the Hilbert–Schmidt Dirichlet integral. The rows r_i of $Q\Phi$ have norm one on $\partial \mathbb{D}_R$ and vanish to orders at least $m_1 = 2$ and $m_i = 1$ for $i \geq 2$. Writing $r_i(z) = \sum_{k \geq m_i} a_{ik}(z/R)^k$, with $a_{ik} \in \mathbb{C}^N$, Parseval's identity and radial integration give

$$\sum_{k \geq m_i} \|a_{ik}\|^2 = 1, \quad \int_{\mathbb{D}_R} |r_i'|^2 dA = \pi \sum_{k \geq m_i} k \|a_{ik}\|^2 \geq \pi m_i.$$

Summing proves the lower bound $\pi(n+1)$. It is attained by $U \operatorname{diag}((z/R)^2, z/R, \dots, z/R)$ when the last $n-1$ columns of $U \in U(n)$ span $\ker L$, appending $N-n$ zero columns if $N > n$. This multiplier optimality leaves the holomorphic realization and its remainder estimates to the next section.

4. UNIFORM HOLOMORPHIC REALIZATION AND DISK OPERATIONS

We realize the trace calculation by holomorphic families and then prove the two disk operations: uniform descent and scalar enlargement. Holomorphic mapping spaces and Stein neighborhoods of graphs provide a qualitative deformation theory [17]. Here the parameter radius and cubic area remainder must remain uniform as the disk and its source radius vary. We obtain this control by lifting to a fixed Stein submanifold of affine space, perturbing there, and projecting back to X .

Assume in this section that X is projective. Choose a projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$; it will be used to represent $\iota \circ f$ by homogeneous coordinates and to construct the lift space $\mathcal{Y}$. The embedding ι is kept fixed as f and R vary, so the constants below may depend on ι .

We use the standard L^p and C^k norms on source regions, with Euclidean area measure. Vector and operator norms are induced by the fixed Hermitian and Euclidean metrics; matrix-valued L^2 estimates use the Hilbert–Schmidt norm.

4.1. Frame and lifting estimates

Uniform realization cannot be obtained by declaring the classes with fixed upper bounds on Area_R and B_R compact up to the source boundary. For $0 < r < 1$ and $\varepsilon > 0$, the functions $h_r : \mathbb{D} \rightarrow \mathbb{C}$ given by

$$h_r(z) := \varepsilon \frac{1 - r^2}{1 - rz}$$

are holomorphic near $\overline{\mathbb{D}}$ and satisfy

$$\int_{\mathbb{D}} |h'_r|^2 dA = \pi \varepsilon^2 r^2, \quad B_1(h_r) \leq \frac{\varepsilon}{2}, \quad |h'_r(1)| = \varepsilon \frac{r(1+r)}{1-r} \rightarrow \infty.$$

Thus bounded area and bounded B_1 do not imply relative compactness in $C^1(\overline{\mathbb{D}})$ (with the Euclidean metric on $\mathbb{C}$); the uniform estimates below must be proved directly.

Boundary spectral factorization and a Green potential bound control a normalized frame. Homogeneous coordinates and Nicolau’s corona theorem then give a uniformly bounded affine lift with bounded Dirichlet integral.

Fix $A_0, B_0 \geq 0$. In the auxiliary results below, f is holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Norms on fixed finite-rank bundles are chosen once and for all.

The following factorization makes the frame in Theorem 4.7 unitary on the boundary, as required by the Ricci-trace identity.

Lemma 4.1. *Let $H : \partial\mathbb{D} \rightarrow \text{Mat}_{n \times n}(\mathbb{C})$ be a smooth positive-definite Hermitian matrix function. There is a map $A : \mathbb{D} \rightarrow \text{GL}_n(\mathbb{C})$, holomorphic in $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$ with invertible boundary values, such that*

$$A(\zeta)^* H(\zeta) A(\zeta) = \text{Id}_n \quad (|\zeta| = 1).$$

Proof. The Wiener–Masani theorem in the smooth boundary form of [3, Theorem 2.1] gives $H = h^* h$, where h and h^{-1} are holomorphic on $\mathbb{D}$ and smooth on $\overline{\mathbb{D}}$. Thus $A := h^{-1}$ has the required boundary normalization. $\square$

For $z \in \mathbb{D}$, let δ_z be the unit point mass at z . The Dirichlet Green function of the unit disk is

$$g(z, w) = \log \left| \frac{1 - \bar{z}w}{z - w} \right|, \quad -\Delta_w g = 2\pi \delta_z.$$

On $\mathbb{D}_R$ we use $g_R(z, w) := g(z/R, w/R)$.

The Green potential controls both the frame and the homogeneous lift. The weighted derivative bound controls its logarithmic singularity.

Lemma 4.2. Let $\varepsilon_f^\omega(w) := |f_z(w)|_\omega^2$ be the area density of f , so that $f^*\omega = \varepsilon_f^\omega dA$ and $\text{Area}_R(f) = \int_{\mathbb{D}_R} \varepsilon_f^\omega dA$. If $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, then

$$\sup_{z \in \mathbb{D}_R} \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w) \leq C_g(A_0, B_0) := C_G(B_0^2 + A_0), \quad (18)$$

where C_G is a universal constant, independent of R and f .

Proof. First reduce to the unit disk. Under the dilation $h(\zeta) := f(R\zeta)$, the identities

$$\text{Area}_1(h) = \text{Area}_R(f), \quad B_1(h) = B_R(f), \quad \varepsilon_h^\omega(\eta) = R^2 \varepsilon_f^\omega(R\eta)$$

show that the hypotheses are unchanged. Since $g_R(R\zeta, R\eta) = g(\zeta, \eta)$, change of variables also preserves the Green-potential integral. We may therefore assume $R = 1$.

Fix $z \in \mathbb{D}$ and set $\delta := 1 - |z|$, its distance to the boundary. Split the integral into the disk $|w - z| < \delta/2$ and its complement in $\mathbb{D}$. On the first region,

$$1 - |w| \geq 1 - |z| - |w - z| > \frac{\delta}{2}, \quad |f_z(w)|_\omega \leq \frac{B_0}{1 - |w|}.$$

Thus $\varepsilon_f^\omega(w) \leq 4B_0^2/\delta^2$. To bound the Green kernel, write

$$1 - \bar{z}w = (1 - |z|^2) + \bar{z}(z - w).$$

Since $1 - |z|^2 = (1 + |z|)\delta \leq 2\delta$, the triangle inequality gives

$$|1 - \bar{z}w| \leq 2\delta + |z - w| \leq \frac{5\delta}{2}, \quad g(z, w) \leq \log \frac{5\delta}{2|z - w|}.$$

Integration in polar coordinates bounds this contribution by

$$\frac{4B_0^2}{\delta^2} 2\pi \int_0^{\delta/2} r \log \frac{5\delta}{2r} dr = \pi B_0^2 \left(\log 5 + \frac{1}{2} \right).$$

On the complementary region $|z - w| \geq \delta/2$, the same triangle inequality instead yields

$$\frac{|1 - \bar{z}w|}{|z - w|} \leq 1 + \frac{2\delta}{|z - w|} \leq 5.$$

Thus $g(z, w) \leq \log 5$ there, and the remaining contribution is at most

$$(\log 5) \int_{\mathbb{D}} \varepsilon_f^\omega dA \leq A_0 \log 5.$$

Adding the two bounds proves (18); one may take the universal constant $C_G := \pi(\log 5 + 1/2)$, independently of z , R , and f . $\square$

We specify the curvature convention in local holomorphic coordinates $w^1, \dots, w^n$. Recall that h is the Gram matrix in (1), with squared norms c^*hc . For a section represented by a coefficient column c , the Chern connection and its curvature are

$$\nabla'c = \partial c + h^{-1}(\partial h)c, \quad \nabla''c = \bar{\partial}c, \quad F^\nabla = \bar{\partial}(h^{-1}\partial h).$$

The coefficient of $dw^\alpha \wedge d\bar{w}^\beta$ in F^∇ is $-\partial_{\bar{w}^\beta}(h^{-1}\partial_{w^\alpha}h)$; the minus sign comes from reversing the order of the two differentials. For $v = \sum_\alpha v^\alpha \partial_{w^\alpha}$ and a tangent vector u with coefficient column c , our curvature tensor is

$$R^\omega(v, \bar{v}, u, \bar{u}) := \sum_{\alpha, \beta} v^\alpha \bar{v}^\beta c^* [-\partial_{\bar{w}^\beta} \partial_{w^\alpha} h + (\partial_{\bar{w}^\beta} h) h^{-1} (\partial_{w^\alpha} h)] c. \quad (19)$$

Thus positive holomorphic bisectional curvature means that (19) is positive for every pair of nonzero tangent vectors v, u at the same point. For a holomorphic disk f , the curvature endomorphism $\mathcal{R}_f^\omega$ is characterized by

$$\langle \mathcal{R}_f^\omega v, v \rangle_\omega = R^\omega(f_z, \bar{f}_z, v, \bar{v}). \quad (20)$$

With the normalizations in (1), a holomorphic section s of f^*T_X satisfies

$$\partial_z \partial_{\bar{z}} |s|_\omega^2 = |\nabla_z s|_\omega^2 - \langle \mathcal{R}_f^\omega s, s \rangle_\omega, \quad f^* \text{Ric}(\omega) = 2 \text{tr}(\mathcal{R}_f^\omega) dA. \quad (21)$$

Choose C_ω so that

$$\|\mathcal{R}_f^\omega(z)\|_{\text{op}} \leq C_\omega \varepsilon_f^\omega(z).$$

Compactness of X gives such a constant.

For a frame $e = (e_1, \dots, e_n)$, write $G = (\langle e_j, e_k \rangle_\omega)$. Matrix inequalities refer to Hermitian quadratic forms. The Green bound and Bochner identity control G throughout the disk and bound the frame's covariant Dirichlet integral. Both estimates enter the lifted-field bounds in Theorem 4.7.

Proposition 4.3. *There are constants $\Gamma \geq 1$ and $D_1 < \infty$, depending only on (X, ω, A_0, B_0) , such that every holomorphic frame e of f^*T_X that is smooth on the closed disk and unitary on its boundary has Gram matrix G satisfying*

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n. \quad (22)$$

Moreover

$$\sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1. \quad (23)$$

In particular, these constants are independent of R , f , and the boundary-unitary frame.

Proof. Berndtsson and Rosay obtain two-sided metric bounds from a bounded curvature potential in [3, Proposition 2.3]. We give the Bochner–Green calculation in terms of A_0, B_0 , including the covariant Dirichlet estimate (23). Set

$$u_G := \text{tr } G = \sum_{a=1}^n |e_a|_\omega^2 > 0, \quad q := \sum_{a=1}^n |\nabla_z e_a|_\omega^2.$$

The curvature bound and (21) give

$$\begin{aligned} \partial_z \partial_{\bar{z}} u_G &= q - \sum_a \langle \mathcal{R}_f^\omega e_a, e_a \rangle_\omega \\ &\geq q - C_\omega \varepsilon_f^\omega \sum_a |e_a|_\omega^2 = q - C_\omega \varepsilon_f^\omega u_G. \end{aligned}$$

Because e_a is holomorphic and the connection preserves the metric, $\partial_z u_G = \sum_a \langle e_a, \nabla_z e_a \rangle_\omega$. Cauchy–Schwarz, applied to the direct sum of the n fibers, gives

$$|\partial_z u_G|^2 \leq \left(\sum_a |e_a|_\omega^2 \right) \left(\sum_a |\nabla_z e_a|_\omega^2 \right) = u_G q.$$

Since $u_G > 0$, logarithmic differentiation now yields

$$\begin{aligned} \partial_z \partial_{\bar{z}} \log u_G &= \frac{\partial_z \partial_{\bar{z}} u_G}{u_G} - \frac{|\partial_z u_G|^2}{u_G^2} \\ &\geq \frac{q}{u_G} - C_\omega \varepsilon_f^\omega - \frac{q}{u_G} = -C_\omega \varepsilon_f^\omega. \end{aligned}$$

Hence

$$\Delta \log u_G \geq -4C_\omega \varepsilon_f^\omega, \quad u_G|_{\partial \mathbb{D}_R} = n.$$

Green representation, with $-\Delta_w g_R(z, w) = 2\pi \delta_z$, reads

$$\log \frac{u_G(z)}{n} = -\frac{1}{2\pi} \int_{\mathbb{D}_R} g_R(z, w) \Delta_w \log u_G(w) dA(w).$$

Since $g_R \geq 0$, Lemma 4.2 yields

$$\log \frac{u_G(z)}{n} \leq \frac{2C_\omega}{\pi} \int_{\mathbb{D}_R} g_R(z, w) \varepsilon_f^\omega(w) dA(w) \leq \frac{2C_\omega C_g(A_0, B_0)}{\pi}.$$

The algebraic dual holomorphic frame $e^a(e_b) = \delta_b^a$ has Gram matrix $G^{-\top} := (G^{-1})^\top$, where $\top$ denotes ordinary transpose. The curvature of the dual bundle is the negative transpose of the original curvature, so its operator norm is bounded by $C_\omega \varepsilon_f^\omega$. Applying the logarithmic trace estimate to the dual frame gives

$$\operatorname{tr} G \leq \Gamma, \quad \operatorname{tr}(G^{-1}) \leq \Gamma, \quad \Gamma := n \exp\left(\frac{2C_\omega C_g(A_0, B_0)}{\pi}\right).$$

For every eigenvalue $\mu > 0$ of G , $\mu \leq \operatorname{tr} G \leq \Gamma$ and $\mu^{-1} \leq \operatorname{tr}(G^{-1}) \leq \Gamma$. These two inequalities give (22).

Summing the Bochner identity gives

$$\partial_z \partial_{\bar{z}} \operatorname{tr} G = \sum_a |\nabla_z e_a|^2 - \operatorname{tr}(\mathcal{R}_f^\omega S), \quad S := \sum_a e_a \otimes e_a^b.$$

Here $e_a^b(v) = \langle e_a, v \rangle_\omega$ denotes metric duality. Since $G = \operatorname{Id}_n$ on $\partial \mathbb{D}_R$, Jacobi's formula gives

$$\partial_\nu \log \det G = \operatorname{tr}(G^{-1} \partial_\nu G) = \partial_\nu \operatorname{tr} G \quad \text{on } \partial \mathbb{D}_R.$$

Consequently Green's formula and (1) give

$$\int_{\mathbb{D}_R} \partial_z \partial_{\bar{z}} \operatorname{tr} G dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu \log \det G ds = -\frac{1}{2} \int_{\mathbb{D}_R} f^* \operatorname{Ric}(\omega).$$

Integrating the Bochner identity and solving for the nonnegative derivative term therefore gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA = \int_{\mathbb{D}_R} \operatorname{tr}(\mathcal{R}_f^\omega S) dA - \frac{1}{2} \int_{\mathbb{D}_R} f^* \operatorname{Ric}(\omega).$$

Now $\operatorname{tr} S \leq n\Gamma$, while the absolute Ricci density is at most $2nC_\omega \varepsilon_f^\omega$, since $f^* \operatorname{Ric} = 2 \operatorname{tr}(\mathcal{R}_f^\omega) dA$. Taking absolute values and using $\operatorname{Area}_R(f) \leq A_0$ gives

$$\sum_a \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq nC_\omega (\Gamma + 1) A_0.$$

Thus $D_1 := nC_\omega (\Gamma + 1) A_0$ is an admissible choice in (23). $\square$

Recall the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ chosen at the beginning of this section. Let ω_{FS} be the Fubini–Study form on its target, with normalization

$$\omega_{\text{FS}} = \frac{i}{2} \partial \bar{\partial} \log \sum_{j=1}^m |Z_j|^2.$$

Choose $c_\iota > 0$ such that $\iota^* \omega_{\text{FS}} \leq c_\iota \omega$; such a constant exists by compactness of X . The displayed Fubini–Study formula is read in homogeneous coordinates $Z = (Z_1, \dots, Z_m) \neq 0$; it is unchanged as a form on projective space when the homogeneous representative is multiplied by a nonvanishing holomorphic function.

We choose homogeneous coordinates with a uniform lower bound for the corona theorem and a Dirichlet bound for the affine lift.

Lemma 4.4. *Every disk satisfying $\operatorname{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$ has a homogeneous lift $\mathbf{s}$ from an open neighborhood of $\overline{\mathbb{D}}_R$ to $\mathbb{C}^m \setminus \{0\}$, holomorphic on that neighborhood, with $[\mathbf{s}] = \iota \circ f$ and $|\mathbf{s}| = 1$ on the boundary. It satisfies*

$$|\mathbf{s}| \leq 1, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \int_{\mathbb{D}_R} f^* \iota^* \omega_{\text{FS}} \leq c_\iota A_0, \quad (24)$$

and

$$|\mathbf{s}(z)| \geq \delta_{\mathbf{s}} > 0, \quad (25)$$

where $\delta_{\mathbf{s}} = \delta_{\mathbf{s}}(X, \omega, \iota, A_0, B_0)$ is independent of R and f .

Proof. The tautological line bundle on $\mathbb{P}^{m-1}$ has fiber $\mathbb{C}Z$ over $[Z]$. Trivialize its pullback by $\iota \circ f$ on a slightly larger disk and choose a nonvanishing section, represented by a lift v in $\mathbb{C}^m \setminus \{0\}$. The positive real-analytic boundary function $|v|$ has a logarithm whose harmonic extension is the real part of a holomorphic function a on a slightly larger disk. Then $\mathbf{s} := e^{-a}v$ has boundary norm one, and the maximum principle gives $|\mathbf{s}| \leq 1$.

The area density of $\iota \circ f$ for the Fubini–Study metric is $\varepsilon_f^{\text{FS}} := |(\iota \circ f)'|_{\text{FS}}^2$, so that $f^* \iota^* \omega_{\text{FS}} = \varepsilon_f^{\text{FS}} dA$. Set $u := \log |\mathbf{s}|^2$. Since $\mathbf{s}$ has no common zero, $\Delta u = 4\varepsilon_f^{\text{FS}}$ and $u = 0$ on the boundary. Also $\Delta |\mathbf{s}|^2 = 4|\mathbf{s}'|^2$, and the normal derivatives of $|\mathbf{s}|^2$ and u agree where $|\mathbf{s}| = 1$. Hence

$$\int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu |\mathbf{s}|^2 ds = \frac{1}{4} \int_{\partial \mathbb{D}_R} \partial_\nu u ds = \int_{\mathbb{D}_R} \varepsilon_f^{\text{FS}} dA,$$

which proves (24). Green representation for u with zero boundary values gives

$$-u(z) = \frac{2}{\pi} \int g_R(z, w) \varepsilon_f^{\text{FS}}(w) dA(w) \leq \frac{2c_l C_g(A_0, B_0)}{\pi}.$$

Thus (25) holds with $\delta_{\mathbf{s}} = \exp(-c_l C_g(A_0, B_0)/\pi)$. $\square$

Write $H^\infty(\mathbb{D})$ for the bounded holomorphic functions with the supremum norm, and let

$$\mathcal{D}(\mathbb{D}) := \left\{ h : \mathbb{D} \rightarrow \mathbb{C} \text{ holomorphic} : \int_{\mathbb{D}} |h'|^2 dA < \infty \right\}.$$

The Dirichlet algebra is

$$\mathfrak{A} := H^\infty(\mathbb{D}) \cap \mathcal{D}(\mathbb{D}), \quad \|h\|_{\mathfrak{A}} := \|h\|_\infty + \left(\int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}.$$

The corona estimate makes the affine lift uniform in the disk. Nicolau’s theorem and the bound at the beginning of its proof [41, pp. 135–136] give the following statement.

Theorem 4.5. *For every $m \geq 1$ and $A, \delta > 0$ there is $C_N(m, A, \delta) < \infty$ such that, whenever $u_1, \dots, u_m \in \mathfrak{A}$ satisfy*

$$\|u_j\|_{\mathfrak{A}} \leq A, \quad \max_j |u_j(z)| \geq \delta \quad (z \in \mathbb{D}),$$

there are $v_1, \dots, v_m \in \mathfrak{A}$ with

$$\sum_j u_j v_j = 1, \quad \|v_j\|_{\mathfrak{A}} \leq C_N(m, A, \delta).$$

The same statement holds on every disk after dilation.

Proof. The case $\delta > A$ is vacuous; assume $\delta \leq A$. For $p = 2$, Nicolau uses the norm

$$\|h\|_{\text{Nic}} := \|h\|_\infty + \left(\frac{1}{\pi} \int_{\mathbb{D}} |h'|^2 dA \right)^{1/2}, \quad \|h\|_{\text{Nic}} \leq \|h\|_{\mathfrak{A}} \leq \sqrt{\pi} \|h\|_{\text{Nic}}.$$

His corona condition is the same maximum lower bound as above [41, p. 135, (1)]. The generators u_j/A therefore have Nicolau norm at most 1 and lower bound δ/A . The estimate at the beginning of his proof [41, p. 136] gives solutions b_j with $\sum_j (u_j/A) b_j = 1$ and $\|b_j\|_{\text{Nic}} \leq K_m(\delta/A)$, where K_m depends only on the indicated data. Setting $v_j := b_j/A$ gives $\sum_j u_j v_j = 1$ and

$$\|v_j\|_{\mathfrak{A}} \leq \frac{\sqrt{\pi}}{A} K_m(\delta/A),$$

which is an admissible choice of $C_N(m, A, \delta)$. For a disk of radius R ,

$$\|h(R \cdot)\|_{\infty, \mathbb{D}} = \|h\|_{\infty, \mathbb{D}_R}, \quad \int_{\mathbb{D}} |(h(R\zeta))'|^2 dA(\zeta) = \int_{\mathbb{D}_R} |h'(z)|^2 dA(z),$$

so the same bound applies after dilation. $\square$

For a normalized lift $\mathbf{s}$ from Lemma 4.4, its coordinates have algebra norm at most $C_s := 1 + \sqrt{c_\iota A_0}$, and at least one has modulus at least $\delta_s/\sqrt{m}$ at every point, because $|\mathbf{s}|^2 \leq m \max_j |\mathbf{s}_j|^2$. Theorem 4.5 gives a holomorphic map $\chi : \mathbb{D}_R \rightarrow (\mathbb{C}^m)^*$, whose components belong to the algebra on $\mathbb{D}_R$ obtained from $\mathfrak{A}$ by dilation, satisfying

$$\chi \cdot \mathbf{s} = 1, \quad \|\chi\|_\infty + \|\chi'\|_2 \leq C_\chi \quad (26)$$

for a constant $C_\chi = C_\chi(X, \omega, \iota, A_0, B_0)$. The dot denotes the bilinear pairing $\chi \cdot \mathbf{s} = \sum_{j=1}^m \chi_j \mathbf{s}_j$, without complex conjugation. The constant absorbs the fixed dimension factors in passing from scalar to vector norms.

To extend the lift past $\overline{\mathbb{D}}_R$, dilate χ radially while keeping $\mathbf{s}$ fixed, then restore their pairing by division. For $0 < \rho < 1$ sufficiently close to 1, set

$$\chi_\rho(z) := \chi(\rho z), \quad \gamma_\rho := \chi_\rho \cdot \mathbf{s}, \quad \hat{\chi} := \frac{\chi_\rho}{\gamma_\rho}.$$

Indeed, $\gamma_\rho - 1 = \chi(\rho \cdot) \cdot (\mathbf{s} - \mathbf{s}(\rho \cdot)) \rightarrow 0$ uniformly on $\overline{\mathbb{D}}_R$, so we may require $|\gamma_\rho| \geq 1/2$ there. The quotient is therefore holomorphic near $\overline{\mathbb{D}}_R$ and satisfies $\hat{\chi} \cdot \mathbf{s} = 1$ and $\|\hat{\chi}\|_\infty \leq 2C_\chi$. Dilation gives $\|\chi'_\rho\|_2 \leq C_\chi$; the product and quotient rules give

$$\begin{aligned} \|\gamma'_\rho\|_2 &\leq C_\chi(1 + \sqrt{c_\iota A_0}), \\ \|\hat{\chi}'\|_2 &\leq 2C_\chi + 4C_\chi^2(1 + \sqrt{c_\iota A_0}) =: D_\chi. \end{aligned} \quad (27)$$

All norms here are on $\mathbb{D}_R$. The bound depends only on the fixed target, embedding, and A_0, B_0 ; the base lift $\mathbf{s}$ is unchanged.

4.2. Uniform lifts and holomorphic variations

The pairs $(\mathbf{s}, \chi)$ lie in a fixed affine model: the equation $\chi \cdot \mathbf{s} = 1$ excludes the cone vertex. A holomorphic right inverse lifts tangent fields to this model. The model depends only on the embedding; the area and weighted-derivative caps determine a compact subset and a retraction near it.

Lemma 4.6. *For the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$ chosen above, there are a closed Stein submanifold $\mathcal{Y} \subset \mathbb{C}^{2m}$, a holomorphic submersion $\pi_X : \mathcal{Y} \rightarrow X$, and a holomorphic bundle map $\mathcal{L} : \pi_X^* T_X \rightarrow T_{\mathcal{Y}}$ such that $d\pi_X \circ \mathcal{L} = \text{Id}$. These objects depend only on the fixed embedding.*

Proof. Consider the affine cone of the embedding,

$$\hat{X} := \{0\} \cup \{\mathbf{s} \in \mathbb{C}^m \setminus \{0\} : [\mathbf{s}] \in \iota(X)\},$$

and set

$$\mathcal{Y} := \{(\mathbf{s}, \chi) \in \hat{X} \times (\mathbb{C}^m)^* : \chi \cdot \mathbf{s} = 1\}, \quad \pi_X(\mathbf{s}, \chi) := \iota^{-1}[\mathbf{s}]. \quad (28)$$

The defining equation excludes the cone vertex. On a chart $\mathbf{s}_j \neq 0$, local coordinates consist of coordinates on X , a nonzero scale factor, and $m - 1$ unconstrained dual coordinates. Hence $\mathcal{Y}$ is a smooth closed affine algebraic variety and therefore Stein, while π_X is a holomorphic submersion.

For the holomorphic splitting, set $V := \mathbb{C}^m$ and let $\ell := \iota^* \mathcal{O}_{\mathbb{P}^{m-1}}(-1)$ be the tautological line bundle on X ; its fiber $\ell_x \subset V$ is the line representing $\iota(x)$. Define

$$\mathcal{U} := \{(x, \chi) \in X \times V^* : \chi|_{\ell_x} \neq 0\}.$$

The map

$$\Theta : \mathcal{Y} \longrightarrow \mathcal{U}, \quad (\mathbf{s}, \chi) \longmapsto (\pi_X(\mathbf{s}, \chi), \chi),$$

is biholomorphic. Indeed, if s_0 is a local nonvanishing holomorphic section of ℓ , then

$$\sigma(x, \chi) := \frac{s_0(x)}{\chi(s_0(x))}$$

is independent of the choice of s_0 , is holomorphic, lies in ℓ_x , and satisfies $\chi(\sigma(x, \chi)) = 1$. Thus

$$\Theta^{-1}(x, \chi) = (\sigma(x, \chi), \chi).$$

Since $\mathcal{U}$ is open in $X \times V^*$, its tangent space at (x, χ) is $T_x X \oplus V^*$. Transporting the vectors $(v, 0)$ by $d\Theta^{-1}$ gives the required holomorphic right inverse

$$\mathcal{L} : \pi_X^* T_X \rightarrow T_Y, \quad \mathcal{L}_y(v) := d(\Theta^{-1})_{\Theta(y)}(v, 0), \quad d\pi_X \circ \mathcal{L} = \text{Id}. \quad (29)$$

□

The normalized coordinates and their corona partners place every disk in a common compact subset of the fixed affine model. The frame estimates then control its lifted tangent fields. We assemble these data before constructing the holomorphic variations.

Theorem 4.7. *Fix the affine model $\mathcal{Y}$, the submersion π_X , and the right inverse $\mathcal{L}$ supplied by Lemma 4.6. Given $A_0, B_0 \geq 0$, one can choose a compact set $K_{A_0, B_0} \Subset \mathcal{Y}$, an open neighborhood Ω of K_{A_0, B_0} in $\mathbb{C}^{2m}$, a holomorphic map $\tilde{\pi}_X : \Omega \rightarrow X$ with $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, and constants*

$$\Gamma, D_1, D_{\tilde{f}}, C_{\theta, \infty}, D_\theta < \infty.$$

These cap-dependent choices depend only on $(X, \omega, \iota, A_0, B_0)$ and are independent of the source radius and the individual disk; the affine model and its splitting depend only on ι . For every $R > 0$ and every map f holomorphic on a neighborhood of $\overline{\mathbb{D}}_R$ with $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$, one can choose a frame e , a lifted disk $\tilde{f}$, and a matrix θ compatibly so that the following properties hold:

- (i) *e is a holomorphic frame of f^*T_X , smooth on the closed disk and unitary on its boundary, whose Gram matrix $G(z) := (\langle e_j(z), e_k(z) \rangle_\omega)_{j,k=1}^n$ satisfies*

$$\Gamma^{-1} \text{Id}_n \leq G \leq \Gamma \text{Id}_n, \quad \sum_{a=1}^n \int_{\mathbb{D}_R} |\nabla_z e_a|^2 dA \leq D_1;$$

- (ii) *$\tilde{f}$ is a map from a neighborhood of $\overline{\mathbb{D}}_R$ to $\mathcal{Y}$, holomorphic on that neighborhood, such that*

$$\pi_X \circ \tilde{f} = f, \quad \tilde{f}(\overline{\mathbb{D}}_R) \subset K_{A_0, B_0}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq D_{\tilde{f}};$$

- (iii) *the frame vectors have holomorphic lifts $\theta_1, \dots, \theta_n : \mathbb{D}_R \rightarrow \mathbb{C}^{2m}$ along $\tilde{f}$. Their column matrix and the lifting identities are*

$$\begin{aligned} \theta &:= (\theta_1, \dots, \theta_n) : \mathbb{D}_R \rightarrow \text{Mat}_{2m \times n}(\mathbb{C}), \\ d(\tilde{\pi}_X)_{\tilde{f}(z)} \theta_a(z) &= e_a(z) \quad (1 \leq a \leq n). \end{aligned} \quad (30)$$

The matrix θ is smooth on the closed disk and satisfies

$$\sup_{z \in \mathbb{D}_R} \|\theta(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq D_\theta.$$

Proof. Fix a disk f satisfying the caps A_0, B_0 .

- (i) Choose $R' > R$ so that $\overline{\mathbb{D}}_{R'}$ lies in the given source neighborhood, and choose a holomorphic frame $\tilde{e}$ of f^*T_X on $\mathbb{D}_{R'}$; such a frame exists by the Oka–Grauert principle [23, Satz 6, p. 270], since the source is contractible and Stein. Restrict its Gram matrix to $\partial\mathbb{D}_R$, rescale by $z = R\zeta$, and apply Lemma 4.1. If $A(\zeta)$ is the resulting factor for the rescaled Gram matrix, set $e(z) := \tilde{e}(z)A(z/R)$. Its boundary Gram matrix is $A(z/R)^* \tilde{G}(z) A(z/R) = \text{Id}_n$. Thus e is holomorphic, smooth on the closed disk, and unitary on its boundary. Proposition 4.3 gives (22) and (23), proving (i).

- (ii) By Lemma 4.4, f has a homogeneous lift $\mathbf{s}$ with

$$|\mathbf{s}| \leq 1, \quad |\mathbf{s}| \geq \delta_{\mathbf{s}}, \quad \int_{\mathbb{D}_R} |\mathbf{s}'|^2 dA \leq c_\iota A_0.$$

Its coordinates satisfy the hypotheses of Theorem 4.5 with algebra-norm bound C_s and corona lower bound $\delta_s/\sqrt{m}$. Thus (26) supplies a vector χ satisfying $\chi \cdot s = 1$, and the dilation construction leading to (27) gives a vector $\hat{\chi}$, holomorphic near the closed disk, with

$$\hat{\chi} \cdot s = 1, \quad \|\hat{\chi}\|_{L^\infty(\mathbb{D}_R)} \leq 2C_\chi, \quad \|\hat{\chi}'\|_{L^2(\mathbb{D}_R)} \leq D_\chi.$$

The identity $\hat{\chi} \cdot s = 1$ shows that $\tilde{f} := (s, \hat{\chi})$ takes values in the fixed space $\mathcal{Y}$ of (28). It satisfies

$$\tilde{f}(\mathbb{D}_R) \subset K_{A_0, B_0} := \mathcal{Y} \cap \{|s| \leq 1, |\chi| \leq 2C_\chi\}, \quad \int_{\mathbb{D}_R} |\tilde{f}'|^2 dA \leq c_\iota A_0 + D_\chi^2 =: D_{\tilde{f}}. \quad (31)$$

The set K_{A_0, B_0} is compact: it is closed and bounded, and $\chi \cdot s = 1$ prevents s from approaching the cone vertex.

Since $\mathcal{Y}$ is a closed Stein submanifold of $\mathbb{C}^{2m}$, the Docquier–Grauert tubular neighborhood theorem [13, Satz 3], in the holomorphic deformation-retraction formulation [18, Theorem 3.1], gives a holomorphic retraction $\mathfrak{r} : \Omega \rightarrow \mathcal{Y} \cap \Omega$ from a neighborhood Ω of K_{A_0, B_0} , equal to the identity on $\mathcal{Y} \cap \Omega$. Its derivatives are bounded on smaller compact neighborhoods of K_{A_0, B_0} . Set $\tilde{\pi}_X := \pi_X \circ \mathfrak{r}$. Since $\pi_X \circ \tilde{f} = f$ and $\tilde{\pi}_X|_{\mathcal{Y} \cap \Omega} = \pi_X$, (31) proves (ii). The choices of K_{A_0, B_0} , Ω , and $\tilde{\pi}_X$ depend only on the fixed target data and the two caps; they are fixed before any individual lifted disk $\tilde{f}$ is chosen. The space $\mathcal{Y}$, the projection π_X , and the splitting $\mathcal{L}$ were already fixed from the embedding alone. (iii) Regard the splitting $\mathcal{L}$ from (29) as a bundle map from $\pi_X^* T_X$ to the ambient trivial bundle $\mathcal{Y} \times \mathbb{C}^{2m}$. In $\nabla \mathcal{L}$, use the pullback Chern connection on $\pi_X^* T_X$ and the flat connection on this ambient bundle. Let

$$L_0 := \sup_{K_{A_0, B_0}} \|\mathcal{L}\|, \quad L_1 := \sup_{K_{A_0, B_0}} \|\nabla \mathcal{L}\|.$$

Set

$$\theta_a(z) := \mathcal{L}_{\tilde{f}(z)} e_a(z).$$

Then $d(\tilde{\pi}_X)_\tilde{f} \theta_a = e_a$. With the preceding connection convention, ambient differentiation gives

$$\theta'_a = \mathcal{L}_{\tilde{f}} \nabla_z e_a + (\nabla_{\tilde{f}'} \mathcal{L}) e_a. \quad (32)$$

The frame and lifted-disk bounds, with $|a + b|^2 \leq 2|a|^2 + 2|b|^2$, give

$$\|\theta\|_{L^\infty, \text{op}} \leq L_0 \sqrt{\Gamma} =: C_{\theta, \infty}, \quad (33)$$

$$\int_{\mathbb{D}_R} \|\theta'\|_{\text{HS}}^2 dA \leq 2L_0^2 D_1 + 2n\Gamma L_1^2 D_{\tilde{f}} =: D_\theta. \quad (34)$$

This proves (iii). $\square$

We now add the lifted variation fields and project to X . Radial regularization extends the fields past the closed disk while keeping the lifted map fixed. The multiplier retains its vanishing orders, and the paired trace converges to the value computed in Section 3. The parameter range and cubic remainder remain uniform throughout this approximation.

The choices have a fixed order. The embedding determines $\mathcal{Y}, \pi_X, \mathcal{L}$, independently of the disks and caps; A_0, B_0 then determine $K_{A_0, B_0}, \Omega, \tilde{\pi}_X$. Together with the multiplier bounds D_Φ, N_Φ , these data determine t_0, C_T before R, f, Φ , or the regularization parameter ρ is chosen. Each disk has its own compatible frame, lift, and lifted fields, subject to the bounds of Theorem 4.7. The choice of ρ may depend on these individual data and on Φ ; for trace approximation it may also depend on the prescribed error ε . The constants t_0, C_T remain unchanged. The holomorphic extension neighborhood may depend on the individual family, Φ , and ρ ; no uniform extension width is asserted.

Theorem 4.8. *Fix (X, ω) , the projective embedding $\iota : X \hookrightarrow \mathbb{P}^{m-1}$, and the bounds $A_0, B_0 \geq 0$ as in Theorem 4.7. Let $D_\Phi \geq 0$, and let $N_\Phi \geq 1$ be an integer. There are constants $t_0 > 0$ and $C_T < \infty$, depending only on $(X, \omega, \iota, A_0, B_0, D_\Phi, N_\Phi)$, with the following property.*

Let $R > 0$, and let $f : \mathbb{D}_R \rightarrow X$ extend holomorphically to a neighborhood of $\overline{\mathbb{D}}_R$ and satisfy $\text{Area}_R(f) \leq A_0$ and $B_R(f) \leq B_0$. Use the fixed K_{A_0, B_0} , Ω , and $\tilde{\pi}_X$ from Theorem 4.7, and choose any compatible disk-dependent triple $e, \tilde{f}, \theta$ supplied by that theorem. Suppose that $\Phi = (\Phi_1, \dots, \Phi_{N_\Phi}) : \mathbb{D}_R \rightarrow \text{Mat}_{n \times N_\Phi}(\mathbb{C})$ extends holomorphically to an open neighborhood of $\mathbb{D}_R$ and satisfies

$$\sup_{z \in \overline{\mathbb{D}}_R} \|\Phi(z)\|_{\text{op}} \leq 1, \quad \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA \leq D_\Phi,$$

where the column Φ_a determines the section $e\Phi_a = \sum_{j=1}^n e_j \Phi_{ja}$ of f^*T_X . The constants t_0 and C_T are independent of R, f, Φ and of this compatible triple.

For each $0 < \rho < 1$, write

$$\Xi_\rho(z) := \theta(\rho z)\Phi(z), \quad (\Xi_\rho)_a(z) := \sum_{j=1}^n \theta_j(\rho z)\Phi_{ja}(z).$$

(i) Holomorphic realization. For every $0 < \rho < 1$, each column defines the family

$$f_{a,t}^\rho(z) := \tilde{\pi}_X(\tilde{f}(z) + t(\Xi_\rho)_a(z)), \quad t \in \mathbb{C}, \quad |t| \leq t_0, \quad 1 \leq a \leq N_\Phi.$$

All these families are jointly holomorphic on one open neighborhood of $\overline{\mathbb{D}}_R \times \{t \in \mathbb{C} : |t| \leq t_0\}$; for every $0 < \rho < 1$ that neighborhood may depend on the selected disk-dependent triple, Φ , and ρ , but not on the column a .

(ii) Uniform cubic remainder. For every $0 < \rho < 1$, $1 \leq a \leq N_\Phi$, and $\tau \in \{1, i\}$, write $t = \tau s$ with $s \in \mathbb{R}$ and set

$$\ell_{a,\tau}^\rho := \left. \frac{d}{ds} \right|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho), \quad H_{a,\tau}^\rho := \left. \frac{d^2}{ds^2} \right|_{s=0} \text{Area}_R(f_{a,\tau s}^\rho).$$

For $|s| \leq t_0$,

$$\text{Area}_R(f_{a,\tau s}^\rho) \leq \text{Area}_R(f) + \ell_{a,\tau}^\rho s + \frac{1}{2} H_{a,\tau}^\rho s^2 + C_T |s|^3 \quad (35)$$

with the same constant C_T for both choices of τ .

(iii) Trace approximation. If Φ is boundary coisometric, meaning $\Phi\Phi^* = \text{Id}_n$ on $\partial\mathbb{D}_R$, then for every $\varepsilon > 0$ there is $0 < \rho_\varepsilon < 1$, depending only on the individual data displayed above and on ε , such that

$$\left| \sum_{a=1}^{N_\Phi} (H_{a,1}^{\rho_\varepsilon} + H_{a,i}^{\rho_\varepsilon}) - \left(4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega) \right) \right| \leq \varepsilon. \quad (36)$$

Only this conclusion requires choosing ρ_ε close to 1; t_0 and C_T remain independent of ε and ρ_ε .

Proof. Use the fixed cap-dependent ambient data and the selected compatible triple from Theorem 4.7. Set

$$\delta_0 := \min\{1, \text{dist}(K_{A_0, B_0}, \mathbb{C}^{2m} \setminus \Omega)\} > 0,$$

and choose a compact $K' \Subset \Omega$ whose interior contains the closed Euclidean $3\delta_0/4$ -thickening of K_{A_0, B_0} . In the standard coordinate frame of $\mathbb{C}^{2m}$, let g_0 be the positive-semidefinite Hermitian coefficient matrix of $\tilde{\pi}_X^* \omega$ on K' , so

$$v^* g_0(w) v = |d(\tilde{\pi}_X)_w(v)|_\omega^2$$

for a coefficient column v . Only smooth coefficient bounds are used; the pullback form may be degenerate along the fibers. Take $C_{g,3}$ to bound the coefficients and their derivatives through order three on K' .

Take a multiplier Φ satisfying the stated bounds and, for $0 < \rho < 1$, set $\Xi_\rho := \theta(\rho \cdot) \Phi$. Dilation gives $\int_{\mathbb{D}_R} |\rho \theta'(\rho z)|^2 dA(z) = \int_{\mathbb{D}_{\rho R}} |\theta'(w)|^2 dA(w) \leq D_\theta$. Use this identity, $\|\Phi\|_\infty \leq 1$, and $(\theta(\rho z) \Phi)' = \rho \theta'(\rho z) \Phi + \theta(\rho z) \Phi'$ to obtain

$$\sup_{z \in \mathbb{D}_R} \|\Xi_\rho(z)\|_{\text{op}} \leq C_{\theta, \infty}, \quad \int_{\mathbb{D}_R} \|\Xi'_\rho\|_{\text{HS}}^2 dA \leq 2D_\theta + 2C_{\theta, \infty}^2 D_\Phi =: D_\Xi. \quad (37)$$

Choose

$$t_0 := \min \left\{ 1, \frac{\delta_0}{4 \max\{1, C_{\theta, \infty}\}} \right\}.$$

For $|t| \leq 2t_0$, the displacement of each column perturbation is at most $2t_0 C_{\theta, \infty} \leq \delta_0/2$. Thus $\tilde{f} + t(\Xi_\rho)_a$ stays in K' , uniformly in the disk and the column.

(i) Fix a column and write $\Xi := (\Xi_\rho)_a$. The function $\theta(\rho z)$ is holomorphic for $|z| < R/\rho$; since $\tilde{f}$ and Φ extend past the closed disk, so does $\tilde{f} + t\Xi$. On the closed disk and for $|t| \leq t_0$ its image lies in the closed $\delta_0/2$ -thickening of K_{A_0, B_0} , which is compactly contained in $\text{int } K' \Subset \Omega$. Compactness gives a neighborhood of $\mathbb{D}_R \times \{|t| \leq t_0\}$ on which the image remains in Ω , so the projected family is jointly holomorphic there. Intersect these neighborhoods over the finitely many columns.

(ii) In ambient coordinates write $w(t) := \tilde{f} + t\Xi$ and $v(t) := \tilde{f}' + t\Xi'$. The area density is $v(t)^* g_0(w(t)) v(t)$. Along either real parameter axis $t = \tau s$, with $\tau \in \{1, i\}$, the functions $w(\tau s)$ and $v(\tau s)$ are affine in s , with first derivatives $\tau\Xi$ and $\tau\Xi'$. For fixed z , put $G(s) := g_0(w(\tau s))$, $V(s) := v(\tau s)$, and $Q := \tau\Xi'$. Ordinary differentiation in the real variable s gives

$$\frac{d^3}{ds^3} (V^* G V) = V^* G''' V + 3Q^* G'' V + 3V^* G'' Q + 6Q^* G' Q.$$

The chain rule bounds $|G^{(k)}(s)|$ by $C_{g,3} |\Xi|^k$ for $1 \leq k \leq 3$, up to fixed dimension factors. The bounds $\|\Xi\|_\infty \leq C_{\theta, \infty}$ and $|t| \leq t_0$ consequently give the integrable estimate

$$\left| \frac{d^3}{ds^3} (v(\tau s)^* g_0(w(\tau s)) v(\tau s)) \right| \leq C(C_{g,3}, C_{\theta, \infty}, t_0) (|\tilde{f}'|^2 + |\Xi'|^2).$$

Integrating and applying (31) and (37) give a uniform constant C_T . Taylor's theorem with remainder then proves (35) simultaneously for the two real parameter axes.

(iii) The tangent fields of the actual families satisfy, for each fixed f ,

$$\xi_{\rho, a} := d(\tilde{\pi}_X)_{\tilde{f}}(\theta(\rho \cdot) \Phi_a) \longrightarrow e\Phi_a \quad \text{in } C^1(\overline{\mathbb{D}}_R) \quad (\rho \uparrow 1),$$

because $\theta(\rho \cdot) \rightarrow \theta$ in the same norm. If Φ is boundary coisometric, Lemma 3.3, continuity of (12), and Proposition 3.4 therefore give

$$\sum_a \mathcal{H}_{R, f}(\xi_{\rho, a}) \longrightarrow 4 \int_{\mathbb{D}_R} \|\Phi'\|_{\text{HS}}^2 dA - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

The summand is $H_{a,1}^\rho + H_{a,i}^\rho$ by Lemma 3.3. Choose ρ_ε close enough to one to obtain the prescribed error. Parts (i)–(ii) and (37) hold for every $\rho \in (0, 1)$ with the same t_0, C_T . Their dependence is only on the fixed target data and A_0, B_0, D_Φ, N_Φ . $\square$

4.3. Uniform descent and enlargement

When $\text{Ric}(\omega) \geq \kappa \omega$ and $\text{Area}_R(f) > A_*$, the trace is negative. Selecting one real direction and the sign of its parameter gives descent; the common parameter radius and Taylor remainder give a uniform step size under fixed area and weighted-derivative bounds.

Theorem 4.9. *Let (X, ω) be a compact Kähler n -fold with $\text{Ric}(\omega) \geq \kappa\omega$ for some $\kappa > 0$, and set $A_* = 2\pi(n+1)/\kappa$ as in (3). Fix the point $x_0 \in X$ and the nonzero complex-linear covector $\lambda \in T_{x_0}^*X$ from (14). For every $A_0, B_0, \delta > 0$ there is $\sigma > 0$ with the following property. For every $R > 0$ and every holomorphic map $f : U \rightarrow X$, where $U \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_R \Subset U$, satisfying*

$$f(0) = x_0, \quad \lambda(f_z(0)) = 1, \quad \text{Area}_R(f) \leq A_0, \quad B_R(f) \leq B_0, \quad \text{Area}_R(f) \geq A_* + \delta,$$

there is a holomorphic map $g : U' \rightarrow X$, with U' an open neighborhood of $\overline{\mathbb{D}}_R$, such that

$$g(0) = x_0, \quad \lambda(g_z(0)) = 1, \quad \text{Area}_R(g) \leq \text{Area}_R(f) - \sigma.$$

Proof. For the multiplier (15), one has $N_\Phi = n$, $D_\Phi = \pi(n+1)$, and, by (17),

$$4\pi(n+1) - 2\kappa \text{Area}_R(f) \leq -2\kappa\delta.$$

Apply Theorem 4.8 with trace error $\varepsilon = \kappa\delta$, choose the value ρ_ε supplied by part (iii), and abbreviate $f_{a,t} = f_{a,t}^{\rho_\varepsilon}$, $\ell_{a,\tau} = \ell_{a,\tau}^{\rho_\varepsilon}$, and $H_{a,\tau} = H_{a,\tau}^{\rho_\varepsilon}$. For every column a and every $|t| \leq t_0$, the identities $\Phi_a(0) = 0$ and $d(\tilde{\pi}_X)_{\tilde{f}(0)}\theta(0) = e(0)$ give the exact interpolation formula

$$\begin{aligned} f_{a,t}(0) &= f(0), \\ (f_{a,t})_z(0) &= f_z(0) + t e(0) \Phi'_a(0), \\ \lambda((f_{a,t})_z(0)) &= \lambda(f_z(0)) + t \lambda(e(0) \Phi'_a(0)) = 1. \end{aligned} \tag{38}$$

Indeed, the ambient perturbation at $z = 0$ equals $\tilde{f}(0)$ for every t , so the chain rule always evaluates $d\tilde{\pi}_X$ at the same point $\tilde{f}(0)$. Moreover, the product rule gives $(\Xi_{\rho_\varepsilon})'_a(0) = \theta(0) \Phi'_a(0)$, so radial regularization does not alter the derivative of the increment at the origin. Formula (38) fixes the displayed scalar component, not the entire first jet. For this choice, part (iii) and the preceding trace bound give $\sum_a (H_{a,1} + H_{a,i}) \leq -\kappa\delta$. One of the $2n$ real axes therefore has second area derivative $H_{a,\tau} \leq -\kappa\delta/(2n)$. Choose the sign of s so that $\ell_{a,\tau}s \leq 0$. Then (35) gives

$$\text{Area}_R(f_{a,\tau s}) - \text{Area}_R(f) \leq -\frac{\kappa\delta}{4n} s^2 + C_T |s|^3.$$

Set

$$t_* := \min \left\{ \frac{t_0}{2}, \frac{\kappa\delta}{8n \max\{C_T, 1\}} \right\}, \quad \sigma := \frac{\kappa\delta}{8n} t_*^2.$$

Taking $s \in \{t_*, -t_*\}$ with the chosen sign and $g = f_{a,\tau s}$ proves the assertion. The constants t_0, C_T , hence t_* and σ , are independent of R, f , and ρ . $\square$

The same trace cost and Taylor bounds apply when the covector is $\lambda_{f(0)}^F$. This gives the relative descent used in continuation, with constants independent of the center in F .

Proof of Proposition 2.1. For descent, use the multiplier (15) with $L_f = \lambda_{f(0)}^F \circ e(0)$. It has

$$\Phi_f(0) = 0, \quad L_f \Phi'_f(0) = 0, \quad \int_{\mathbb{D}_R} \|\Phi'_f\|_{\text{HS}}^2 dA = \pi(n+1).$$

By (38), each variation fixes $f(0)$ and the scalar $\lambda_{f(0)}^F(f_z(0)) = 1$. The choice of t_* and σ in the proof of Theorem 4.9 uses only the trace bound and the uniform Taylor estimate. In the present family, the parameter radius and cubic remainder in Theorem 4.8 depend only on A_0, B_0 and the fixed target data. Hence the same area drop works for all $f(0) \in F$. $\square$

For scalar enlargement, it suffices to lift one finite-area disk and perturb its lift in one constant ambient direction.

Proof of Lemma 2.2. By Lemma 2.7, $B_R(h) < \infty$. Choose $s_j \uparrow R$. Each restriction $h|_{\mathbb{D}_{s_j}}$ extends holomorphically past its closed disk and has the common bounds

$$\text{Area}_{s_j}(h) \leq A_0 := \text{Area}_R(h), \quad B_{s_j}(h) \leq B_0 := B_R(h).$$

Apply Theorem 4.7(ii) with these caps. It supplies one compact set $K = K_{A_0, B_0} \Subset \mathcal{Y}$, one ambient neighborhood $\Omega \subset \mathbb{C}^{2m}$ of K , and one holomorphic projection $\Pi = \tilde{\pi}_X : \Omega \rightarrow X$ satisfying $\Pi|_{\mathcal{Y} \cap \Omega} = \pi_X$. The corresponding lifts H_j obey

$$H_j(\overline{\mathbb{D}_{s_j}}) \subset K, \quad \Pi \circ H_j = h \quad \text{on } \mathbb{D}_{s_j}, \quad \int_{\mathbb{D}_{s_j}} |H'_j|^2 dA \leq D_{\tilde{f}},$$

where the last constant is independent of j .

The compact range makes the coordinate functions of H_j a normal family on every compact subdisk of $\mathbb{D}_R$. A diagonal subsequence therefore converges locally uniformly, together with its derivatives, to a holomorphic map $H : \mathbb{D}_R \rightarrow \mathbb{C}^{2m}$. Since K is closed, $H(\mathbb{D}_R) \subset K$, and continuity gives $\Pi \circ H = h$. For every $s < R$, convergence of derivatives on $\overline{\mathbb{D}_s}$ gives $\int_{\mathbb{D}_s} |H'|^2 dA \leq D_{\tilde{f}}$. Exhausting $\mathbb{D}_R$ yields

$$H(\mathbb{D}_R) \subset K, \quad \Pi \circ H = h, \quad \int_{\mathbb{D}_R} |H'|^2 dA \leq D_{\tilde{f}}. \quad (39)$$

No compatibility between the separately chosen lifts is required.

Set $x = h(0)$, $y = H(0)$, and take the constant ambient vector $v = H'(0)$. Since $\Pi \circ H = h$, this choice satisfies $\lambda_x^F(d\Pi_y v) = 1$. Choose $\epsilon > 0$ such that the closed ambient ϵ -neighborhood of K is compact and contained in Ω . Since $R < \infty$, there is $t_0 > 0$ with $t_0 R |v| < \epsilon$. For $0 \leq t < t_0$ define

$$h_t(z) := \Pi(H(z) + t z v), \quad z \in \mathbb{D}_R.$$

This is holomorphic on the whole open disk and fixes x . At zero its derivative satisfies the exact identity

$$(h_t)_z(0) = d\Pi_y((1+t)H'(0)) = (1+t)h_z(0), \quad \lambda_x^F((h_t)_z(0)) = 1+t. \quad (40)$$

The pullback Hermitian form $\Pi^* \omega$ has bounded coefficients on the chosen compact ambient neighborhood. After decreasing t_0 to at most 1, the area density of h_t is consequently bounded by

$$C(|H'(z)|^2 + |v|^2)$$

for a constant C independent of z and t . This function is integrable on $\mathbb{D}_R$, by (39) and $R < \infty$. The densities converge pointwise to the density of h as $t \downarrow 0$, so dominated convergence gives $\text{Area}_R(h_t) \rightarrow \text{Area}_R(h)$. Finally, source dilation gives

$$G_t(0) = h(0), \quad \lambda_{h(0)}^F((G_t)_z(0)) = 1, \quad \text{Area}_{(1+t)R}(G_t) = \text{Area}_R(h_t).$$

In fact, $(G_t)_z(0) = h_z(0)$ for this choice of v ; only its scalar component is needed in the continuation argument. Restriction to any radius strictly between R and $(1+t)R$ gives the closed-disk extension required for an admissible competitor. $\square$

5. COMPACTNESS, MARKED VALUES, AND FINITE-AREA DISKS

The main proof uses compactness to obtain a principal map in the original source coordinate and a rational chain retaining two marked values. We obtain both from one fixed-mark exhaustion statement, with a joint area bound for the principal map and the attached spheres. Point selection then gives the weighted derivative bound for an individual finite-area disk. Throughout this section, (X, ω) is compact Kähler; a Ricci lower bound is not required.

5.1. Finite-area removal

Finite area fills the punctures of a principal limit, including infinity when its source is $\mathbb{C}$. We first prove removal so that the compactness statement can include the extended principal map.

With $g_\omega(v, w) = \omega(v, Jw)$, the energy convention in the compactness and removal results used below is

$$\|df\|_{L^2}^2 = \int (|f_x|_{g_\omega}^2 + |f_y|_{g_\omega}^2) dA = 2 \int f^* \omega = 2 \text{Area}(f), \quad (41)$$

because $f_y = Jf_x$. Thus their energy hypotheses are equivalent to our area hypotheses, with the fixed factor 2.

Proof of Theorem 2.4. Finite area gives $\int_{0 < |z| < r} f^* \omega \rightarrow 0$ as $r \downarrow 0$, so every sufficiently small concentric annulus has small energy by (41). Compactness of X supplies completeness of the target metric and uniform continuity of its complex structure. The removal theorem of Ivashkovich and Shevchishin [29, Corollary 3.6] therefore gives a continuous extension at the puncture. In a target chart about the limiting value, the coordinate functions extend holomorphically by the Riemann removable singularity theorem. For an entire map, $\zeta = 1/z$ identifies infinity with a puncture and preserves area, so the same argument applies there. See also [37, Theorem 4.2.1]. $\square$

5.2. Marked compactness on an exhaustion

Spheres obtained by rescaling near energy concentration points go back to Sacks and Uhlenbeck [43, pp. 2–3 and Section 4]; Gromov’s compactness theory records how such components attach [24]. We use the theorem of Ivashkovich and Shevchishin [30]. For precise locators, we refer to its preprint formulation, whose title and numbering differ from the published version: Definition 1.6, Theorem 1.1, and the marked-point construction in Remark 2 following that theorem [29].

The *principal map* is the limit in the original source coordinate and may be constant. A *bubble* is a sphere obtained by rescaling. A *ghost sphere* is a constant component retaining marks or separating nodes; it has at least three special points, namely marks or nodes. The fixed-coordinate root disk has no such stability requirement. The dual graph has a vertex for each component and an edge for each node; here it is a rooted tree. The finite-area extension of an entire principal map is also a sphere and need not be a bubble.

The following proposition combines the unmarked principal limit needed for the area profile with the marked limit needed for incidence. Fixed distinct source marks suffice for both applications. Constant components inside concentration trees must still be retained, since they can carry a marked value or join nonconstant components.

Proposition 5.1. *Let $R_j > 0$ and let $f_j : U_j \rightarrow X$ be holomorphic, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$. Suppose $\sup_j \text{Area}_{R_j}(f_j) < \infty$, and set $A_\infty = \liminf_j \text{Area}_{R_j}(f_j)$. Choose $R_\infty \in (0, \infty]$ with $R_\infty \leq \liminf_j R_j$, interpreting $\mathbb{D}_\infty$ as $\mathbb{C}$. Fix finitely many pairwise distinct points $a_1, \dots, a_m \in \mathbb{D}_{R_\infty}$, allowing $m = 0$. After first passing to a subsequence realizing A_∞ and then to a further subsequence, the following data exist simultaneously.*

- (i) *A finite set $\Sigma_\infty \subset \mathbb{D}_{R_\infty}$ and a holomorphic principal map $f_\infty : \mathbb{D}_{R_\infty} \rightarrow X$ satisfy*

$$f_j \longrightarrow f_\infty \quad \text{smoothly on compact subsets of } \mathbb{D}_{R_\infty} \setminus \Sigma_\infty$$

in the original source coordinate.

- (ii) *Finitely many holomorphic sphere components $u_\alpha : \mathbb{P}^1 \rightarrow X$, possibly constant, form trees attached to the principal root at points of Σ_∞ . Each attached tree contains a nonconstant component; each constant sphere is stable. If N_α is the set of nodes on the component and z_α is the root attachment point of the tree containing that component, there are source reparametrizations, defined near each compact subset of $\mathbb{P}^1 \setminus N_\alpha$ for all large j , with*

$$f_j \circ \varphi_{j,\alpha} \longrightarrow u_\alpha, \quad \varphi_{j,\alpha} \longrightarrow z_\alpha$$

smoothly for the first convergence and locally uniformly for the second, on compact subsets of $\mathbb{P}^1 \setminus N_\alpha$.

- (iii) *Root, spheres, nodes and marks form one nodal map. Writing u_v for its component maps, p_{vw}, p_{wv} for the two points of a node, and p_ℓ on v_ℓ for the limiting ℓ th mark, one has*

$$u_v(p_{vw}) = u_w(p_{wv}), \quad f_j(a_\ell) \longrightarrow u_{v_\ell}(p_\ell).$$

A marked value therefore propagates along any path of constant components to its first nonconstant component.

- (iv) *The areas, counted with the multiplicities of the maps, satisfy*

$$\text{Area}_{R_\infty}(f_\infty) + \sum_{\alpha} \text{Area}(u_\alpha) \leq A_\infty. \quad (42)$$

Proof. Retain a subsequence whose total areas converge to A_∞ . The proof first selects the principal map on the whole exhaustion, then makes one marked extraction on a fixed disk, and finally counts their areas together.

The original-coordinate principal map.

A diagonal weak selection of the area measures on compact subdisks gives a Radon measure μ on $\mathbb{D}_{R_\infty}$. For every continuous compactly supported χ with $0 \leq \chi \leq 1$,

$$\int \chi d\mu = \lim_j \int \chi f_j^* \omega \leq A_\infty,$$

so $\mu(\mathbb{D}_{R_\infty}) \leq A_\infty$ by exhaustion. Fix $p > 2$ and an area threshold $\epsilon_0 > 0$ small enough for the local compactness result [29, Lemma 3.1 and Corollary 3.3], using (41), and set

$$\Sigma_\infty = \{z : \mu(\{z\}) \geq \epsilon_0\}.$$

The mass bound makes this set finite. For $z \notin \Sigma_\infty$, choose a source disk with μ -null boundary and μ -mass less than ϵ_0 . Weak convergence puts its area below that threshold for all large j . The cited result gives strong local $W^{1,p}$ convergence on a smaller disk; Sobolev embedding gives uniform convergence, and target charts and Cauchy estimates give smooth convergence. A diagonal choice on a countable cover gives f_∞ on the complement of Σ_∞ in the original coordinate. Its local definitions agree because they are limits of the same subsequence. Smooth convergence and exhaustion give $\text{Area}(f_\infty) \leq A_\infty$. Theorem 2.4 fills the finitely many punctures without changing area.

One marked extraction and its root.

Choose fixed radii $0 < r < \rho < r' < R_\infty$ such that $\Sigma_\infty \cup \{a_1, \dots, a_m\} \subset \mathbb{D}_r$. Apply the marked compactness theorem on $\mathbb{D}_{r'}$, retaining $\mathbb{D}_\rho$, to the subsequence already chosen. The standard complex structure is fixed on this source disk, so the boundary-annulus nondegeneration condition is automatic [29, Definition 1.7 and Theorem 1.1]. The positive collar between $\mathbb{D}_\rho$ and $\partial\mathbb{D}_{r'}$ keeps this application strictly inside the available domains. The target is compact, and the energy bound follows from (41).

The fixed-domain construction uses the original coordinate away from concentration points and creates sphere components by rescaling shrinking regions at those points [29, proof of Theorem 1.1, Case 1, and the remark following its proof]. Its root coordinate is identified with the original disk: the holomorphic identification agrees with the identity on an outer open set and hence everywhere. Uniqueness of the already selected limit then identifies the root map with f_∞ off Σ_∞ , and holomorphic extension identifies it throughout $\mathbb{D}_\rho$. We extend this root by the global map f_∞ .

The marked construction in Remark 2 following the same theorem retains the marked evaluations, inserting stable constant components when a mark approaches a node in a rescaled coordinate. Nonconstant components can attach only in Σ_∞ , since convergence is smooth elsewhere. Nor can a tree made entirely of constant components attach to the root: if it had $g \geq 1$ vertices and m_z

marks, stability would give

$$3g \leq 2(g-1) + 1 + m_z, \quad m_z \geq g+1 \geq 2.$$

All these marks would have the same original-coordinate limit, contrary to the fixed distinct marks. Thus all attached trees lie over Σ_∞ and contain a nonconstant sphere. The shrinking-region construction gives the reparametrizations and their collapse asserted in (ii), including for internal ghosts. The compactness theorem gives finiteness of the marked tree.

By Definition 1.6 of the cited preprint, all components belong to one continuous nodal limit. Its two branches therefore have equal values at every node. Marked convergence gives the second identity in (iii); repeatedly applying the node identity propagates a marked value through constant components. No derivative at a marked point is asserted to survive.

Joint global area accounting.

Choose a compact subset $Q \subset \mathbb{D}_{R_\infty} \setminus \Sigma_\infty$ and compact subsets $Q_\alpha \subset \mathbb{P}^1 \setminus N_\alpha$ on the sphere components. The fixed extraction represents the Q_α in mutually disjoint shrinking source regions at separated scales. For all large j these regions also avoid Q , even when Q extends outside the retained disk $\mathbb{D}_\rho$, because Q stays a positive distance from Σ_∞ . Taking the sequence limit first gives

$$\int_Q f_\infty^* \omega + \sum_\alpha \int_{Q_\alpha} u_\alpha^* \omega = \lim_j \left(\int_Q f_j^* \omega + \sum_\alpha \int_{\varphi_{j,\alpha}(Q_\alpha)} f_j^* \omega \right) \leq A_\infty.$$

Now exhaust the global principal domain minus Σ_∞ and each sphere minus its nodes. The omitted points have zero area, and all omitted source areas are nonnegative. This proves (42). Constant components contribute zero; multiplicities are retained because we integrate pullbacks by the component maps. $\square$

Taking no marks gives exactly the principal compactness statement used in the continuation argument.

Proof of Proposition 2.3. Apply Proposition 5.1 with $m = 0$ and omit the nonnegative sphere areas in (42). Its principal map and convergence have the required properties. $\square$

5.3. Two marked values and incidence

Bubbling at 0 preserves the marked value but can separate a prescribed tangent direction from the component carrying the area. Fixing even the complete first jet at 0 does not prevent this loss. On the unit disk, with the product Fubini–Study metric, consider

$$\widehat{f}_j(z) := ([jz^2 : 1], [z : 1]) : \mathbb{D} \longrightarrow \mathbb{P}^1 \times \mathbb{P}^1.$$

The areas are uniformly bounded, and in the affine coordinates $([w : 1], [v : 1])$ the complete first jet is fixed:

$$\widehat{f}_j(0) = ([0 : 1], [0 : 1]), \quad (\widehat{f}_j)_z(0) = (0, 1).$$

Away from zero, the principal limit is

$$\widehat{f}_\infty(z) = ([1 : 0], [z : 1]).$$

Its extension has value $([1 : 0], [0 : 1])$ at zero, different from the marked value. Rescaling by $z = \zeta/\sqrt{j}$ instead gives

$$\widehat{f}_j(\zeta/\sqrt{j}) = ([\zeta^2 : 1], [\zeta/\sqrt{j} : 1]) \longrightarrow ([\zeta^2 : 1], [0 : 1]).$$

The prescribed second-factor direction remains on the residual principal component. Correspondingly, $B_1(\widehat{f}_j)/|(\widehat{f}_j)_z(0)|_\omega \rightarrow \infty$. Thus prescribing the first jet at 0 does not ensure that a nonconstant bubble retains that tangent direction.

Marked and nodal values retain incidence. Lemma 2.5 gives a chain joining two compact target sets with controlled total area. It supplies the compactness alternative in Section 2 and the final passage to the sharp area bound.

Proof of Lemma 2.5. Pass first to a subsequence realizing the original lower limit $A_\infty := \liminf_j \text{Area}_{R_j}(f_j)$. Apply Proposition 5.1 with $R_\infty = \infty$ and the fixed marks $0, 1$. It supplies an entire principal map f_∞ and one finite marked tree, with the joint area bound (42). Theorem 2.4 fills infinity, producing the principal sphere. Replace the principal root of the marked limit by this extension and retain its attached sphere trees. Each mark lies either on the principal sphere or on one of these trees. Its marked value lies in K_0 or K_1 because these sets are closed. On the resulting finite tree, take the path between the marked components and omit constant components. Nodal equality gives nonconstant sphere maps $u_1, \dots, u_\ell$ with

$$u_1(\mathbb{P}^1) \cap K_0 \neq \emptyset, \quad u_\ell(\mathbb{P}^1) \cap K_1 \neq \emptyset, \quad u_i(\mathbb{P}^1) \cap u_{i+1}(\mathbb{P}^1) \neq \emptyset \quad (1 \leq i < \ell).$$

There is at least one such component since $K_0 \cap K_1 = \emptyset$. The components on the path are selected from the global principal sphere and the sphere components of the fixed rooted limit. The joint area estimate (42), after omitting unused components of nonnegative area, gives

$$\sum_{i=1}^{\ell} \int_{\mathbb{P}^1} u_i^* \omega \leq A_\infty.$$

The path belongs to the single marked limit supplied by the proposition. Components arising near the moving outer boundary are not needed for its incidence or area estimate. $\square$

5.4. Rescaling and finite-area weighted derivatives

Brody reparametrization [5, Lemma 2.1], in the affine form described by Duval [16, Section 1], turns divergence of B_R into a nonconstant entire limit. For a single finite-area disk, the positive rescaled area cannot persist in shrinking boundary collars; this gives Lemma 2.7.

Lemma 5.2. *Let $f_j : U_j \rightarrow X$ be holomorphic maps, where $U_j \subset \mathbb{C}$ is open and $\overline{\mathbb{D}}_{R_j} \Subset U_j$, and suppose $B_{R_j}(f_j) \rightarrow \infty$. Then after affine source rescaling, a subsequence converges locally smoothly on $\mathbb{C}$ to a nonconstant entire map $u : \mathbb{C} \rightarrow X$. If in addition $\sup_j \text{Area}_{R_j}(f_j) < \infty$, then*

$$\text{Area}(u) \leq \liminf_j \text{Area}_{R_j}(f_j).$$

Proof. Under the additional area hypothesis, first pass to a subsequence for which $\text{Area}_{R_j}(f_j)$ converges to the lower limit of the original area sequence. In either case, choose w_j maximizing $(R_j - |z|)|(f_j)_z(z)|$, and set $d_j := R_j - |w_j|$, $\Lambda_j := |(f_j)_z(w_j)|$. Then

$$d_j \Lambda_j = B_{R_j}(f_j) \rightarrow \infty, \quad |(f_j)_z(z)| \leq 2\Lambda_j \quad (|z - w_j| < d_j/2).$$

Indeed, $|z - w_j| < d_j/2$ implies $R_j - |z| > d_j/2$. Maximality then gives $(d_j/2)|(f_j)_z(z)| \leq d_j \Lambda_j$, which is the second bound. The rescaled map

$$u_j(\zeta) := f_j(w_j + \zeta/\Lambda_j), \quad |\zeta| < \frac{1}{2}d_j \Lambda_j,$$

satisfies $|(u_j)_\zeta|_\omega \leq 2$ and $|(u_j)_\zeta(0)|_\omega = 1$. The source radii $d_j \Lambda_j/2$ tend to infinity. On every fixed compact source disk, the derivative bound gives equicontinuity, and compactness of X gives a uniformly convergent subsequence. A diagonal selection and Cauchy estimates in local target charts give a smooth limit $u : \mathbb{C} \rightarrow X$ with $|u_\zeta(0)|_\omega = 1$; hence u is nonconstant. Under the additional area hypothesis, conformal invariance gives, for each fixed $S > 0$,

$$\text{Area}_S(u) = \lim_j \text{Area}_S(u_j) \leq \lim_j \text{Area}_{R_j}(f_j) = \liminf_j \text{Area}_{R_j}(f_j).$$

Letting $S \rightarrow \infty$ proves the area bound. $\square$

Proof of Lemma 2.7. Suppose $B_R(h) = \infty$. As $s \uparrow R$, the quantities $B_s(h)$ increase to $B_R(h)$: for each fixed $z \in \mathbb{D}_R$, $(s - |z|)|h_z(z)|_\omega$ tends to $(R - |z|)|h_z(z)|_\omega$ once $s > |z|$, and the suprema increase. Choose $s_j \uparrow R$ with $B_{s_j}(h) \rightarrow \infty$, and apply the point-selection construction of Lemma 5.2 to $h|_{\mathbb{D}_{s_j}}$.

These restrictions extend holomorphically past their closed disks. The construction gives centers $w_j \in \mathbb{D}_{s_j}$ and numbers $\Lambda_j = |h_z(w_j)|_\omega$ such that

$$(s_j - |w_j|)\Lambda_j \longrightarrow \infty, \quad u_j(\zeta) := h(w_j + \zeta/\Lambda_j) \longrightarrow u(\zeta)$$

smoothly on compact subsets of $\mathbb{C}$, where $|u_\zeta(0)|_\omega = 1$. Because $s_j \leq R$, we have $\Lambda_j \rightarrow \infty$. The derivative of the fixed map h is bounded on each compact subdisk, so $|w_j| \rightarrow R$.

Choose $S > 0$ with $\text{Area}_S(u) > 0$. For all large j , the disk $\mathbb{D}_{S/\Lambda_j}(w_j)$ lies in $\mathbb{D}_{s_j}$, and conformal invariance and smooth convergence give

$$\int_{\mathbb{D}_{S/\Lambda_j}(w_j)} h^* \omega = \text{Area}_S(u_j) \longrightarrow \text{Area}_S(u) > 0.$$

On the other hand, $|w_j| \rightarrow R$ and $S/\Lambda_j \rightarrow 0$ place these disks in arbitrarily thin boundary collars of $\mathbb{D}_R$. Since $\text{Area}_R(h) < \infty$, $\int_{\mathbb{D}_R \setminus \overline{\mathbb{D}_s}} h^* \omega \rightarrow 0$ as $s \uparrow R$. This contradicts the preceding positive limit. $\square$

6. RATIONAL CONNECTEDNESS AND AMPLE TANGENT BUNDLES

The analytic construction supplies the curves needed for two classical algebraic arguments. We recall only the conditions that allow those arguments to apply; for the general theory of rational curves, see Kollár [34] and Debarre [11].

Proof of Theorem 1.3. Choose a positive-Ricci metric as in Proposition 3.1. Let $q : X \dashrightarrow Y$ be any quasifibration with $\dim Y > 0$, represented on dense open subsets by a proper morphism $q : X^\circ \rightarrow Y^\circ$ with connected fibers. After shrinking Y° , generic smoothness makes q smooth [28, III, Corollary 10.7]. For a general $y \in Y^\circ$, the full fiber $F = q^{-1}(y)$ is therefore smooth and compact. Theorem 1.1, applied to q near F , gives a rational curve meeting F but not contained in it. By Campana's criterion [6, Proposition 2.7], this property for every such quasifibration implies rational chain connectedness. Here Campana's term "rationnellement connexe" refers to rational chains. The characteristic-zero smoothing theorem then gives rational connectedness in the sense used here [2]. $\square$

Proof of Theorem 1.4. For $n = 1$, ampleness gives $\deg T_X = 2 - 2g(X) > 0$, so $X \simeq \mathbb{P}^1$. Assume $n \geq 2$. Since T_X is ample, its determinant $-K_X$ is ample [26]. Thus X is Fano, and Theorem 1.2 supplies a rational curve. Starting with this curve, the two-point degree-reduction argument of Mori [39] gives a rational curve of anticanonical degree at most $n + 1$; we use precisely the first step in the proof of [12, Section 5, Theorem 3]. That step degenerates an already existing rational curve and decreases its positive integral degree. The genus-zero stable-map compactification [22, Theorem 1] gives a modern setting for this degeneration.

Let $f : \mathbb{P}^1 \rightarrow X$ be the normalization of the resulting curve. Ampleness gives $f^*T_X = \bigoplus_{i=1}^n \mathcal{O}(a_i)$ with every $a_i \geq 1$. The nonzero differential $\mathcal{O}(2) \rightarrow f^*T_X$ forces some $a_i \geq 2$. Consequently its anticanonical degree is at least $n + 1$, and equality gives

$$f^*T_X \simeq \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus(n-1)}.$$

The same differential is then nowhere zero. These are the standard minimal-curve conditions in [12, Section 5, Corollary (a),(c)]. The remaining classification argument of Mori, starting from such a curve, identifies X with $\mathbb{P}^n$; see [12, Section 6]. $\square$

The degree reduction and classification used here take place over $\mathbb{C}$. In the classical proof, the positive-characteristic production of the first curve and the subsequent spreading out are the second and third steps of [12, Section 5, Theorem 3]; our analytic construction replaces those steps. Related numerical characterizations of projective space were developed by Cho, Miyaoka, and Shepherd-Barron [9], with a shorter proof by Kebekus [32].

7. COMPARISON WITH MORI’S METHOD AND NUMERICAL POSITIVITY

For the surrounding birational theory, see Kollár and Mori [36]; Debarre [10, Sections 2–3] gives an account of bend-and-break and rational quotients.

The analytic construction and Mori’s method begin with different ways of producing room for deformation. In Mori’s argument, Frobenius amplifies the anticanonical degree of a map while keeping the genus of its source fixed. Deformation theory then produces a family to which bend-and-break applies; spreading out transfers the resulting rational curves to characteristic zero [39, 38]. Here the source is an expanding disk. Ricci curvature supplies a decreasing area direction, scalar enlargement and source dilation restore the normalized derivative on a larger disk, and finite-area compactness supplies a sphere. The construction of the first rational curve thus takes place over $\mathbb{C}$. The applications then use rational quotients and characteristic-zero minimal-family arguments.

The advantages of this approach concern the construction itself. It links entire curves to rational curves through a precise growth condition, gives the metric bound $2\pi(n+1)/\kappa$, and retains enough incidence information to produce a curve escaping a compact fiber. The cost is substantial analytic preparation: bounded lifts, uniform second-variation estimates, and marked compactness. Compactness also means that the construction is not an effective algorithm for writing equations of the resulting curve. Its present scope is narrower than that of the numerical methods of Mori theory.

The essential limitation is the positivity hypothesis. The trace identity (13), with the multiplier cost (16), gives

$$4\pi(n+1) - 2 \int_{\mathbb{D}_R} f^* \text{Ric}(\omega).$$

We use the pointwise inequality $\text{Ric}(\omega) \geq \kappa\omega$ to make this expression negative whenever the area exceeds A_* , uniformly over the disks encountered in the construction. This is positivity of the Ricci form in every tangent direction, not merely positivity of scalar curvature. By contrast, the Miyaoka–Mori criterion starts from the numerical inequality $-K_X \cdot C > 0$ for one curve C and produces rational curves through its points [38, Theorem 5]. The recent optimal degree refinement of Jovinelly, Lehmann, and Riedl retains this numerical hypothesis and the Frobenius step [31, Theorem 1.1 and its proof]. The degree of a closed curve controls one integrated curvature quantity; it gives no comparable lower bound along all the open disks and deformations used here.

This distinction also matters for the uniruledness criterion of Boucksom, Demailly, Păun, and Peternell [4]. Their cone duality produces a covering family (C_t) with $K_X \cdot C_t < 0$ when K_X is not pseudo-effective. The deduction of uniruledness then explicitly uses Miyaoka–Mori [38, Theorem 5]. Thus cone duality and the production of rational curves are separate steps. Our present argument does not replace the latter step under this numerical hypothesis; a single such curve need not itself belong to a covering family. Campana also emphasizes the role of positive characteristic in this numerical approach [7, Theorems 3.4–3.5 and the following paragraph].

Remark 7.1. Let X be a smooth complex projective variety, let $f : C \rightarrow X$ be a nonconstant map from a smooth connected projective curve, and let $p \in C$. Can one construct a rational curve through $f(p)$ by holomorphic disk deformations and finite-area compactness under the sole assumption

$$-K_X \cdot f_*[C] > 0?$$

The existence conclusion is supplied by Miyaoka–Mori [38]; the problem is to obtain it by an analytic construction in characteristic zero. The missing ingredient is a replacement for uniform Ricci descent derived from this numerical positivity, together with area control through continuation. Such a construction for a covering family would provide an analytic replacement for the curve-production step in the BDPP uniruledness argument [4].

The numerical problem above is related to Yang’s proposed analytic construction of rational curves under RC-positive curvature [47]. Ou extended the uniruledness criterion to compact Kähler manifolds [42], with a subsequent analytic approach by Cao and Păun [8]. These proofs still use

the projective criterion of Boucksom, Demailly, Păun, and Peternell [4], and hence the positive-characteristic methods of Miyaoka–Mori [38]; replacing this decisive step analytically remains a central challenge.

ACKNOWLEDGMENTS

We are grateful to Franc Forstnerič for introducing us to Oka geometry and for his encouragement.

S.-Y. Xie acknowledges partial support from the National Key R&D Program of China under Grants No. 2023YFA1010500 and No. 2021YFA1003100, and from the National Natural Science Foundation of China under Grants No. 12288201 and No. 12471081, as well as support from the Xiaomi Young Talents Program.

AI USE DISCLOSURE

The central mathematical ideas and overall approach are due to the authors. AI tools assisted with manuscript preparation and revision. The authors take full responsibility for the paper’s content and the accuracy of its references.

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